Thermochemical Modeling of a Phenolic Liner in a Solid Rocket Motor Using ANSYS Mechanical

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Contents

Purpose of this document. *This report describes the transient thermal model for phenolic-liner pyrolysis and its extension, now implemented in ANSYS Mechanical, to represent outgassing, blowing, and discrete recession by element deactivation. The Simulation 2 section presents the numerical architecture and an initial performance assessment frozen at $t = 6.15\,\mathrm{s}$ out of the planned $7\,\mathrm{s}$. These results remain provisional until the computation is complete. Fluent and propellant combustion are outside the scope of this study. Values marked* provisional *must be replaced or validated using measurements of the material actually manufactured before any design or safety decision is made.*

1 Context: Solid Rocket Motor and Casing Layup

1.1 Physical Scope

The system under study is a solid-propellant rocket motor whose casing is modeled as four contacting coaxial tubes: an inner phenolic layer, a thin epoxy layer, a second phenolic layer, and an aluminum shell. The geometry file contains the bodies `Phenolic0`, `Epoxy`, `Phenolic1`, and `Alu`, which correspond respectively to the Mechanical named selections `phe0`, `epo`, `phe1`, and `alu`. Mechanical nevertheless meshes only an axially truncated portion of this baseline model: the radial layup is preserved, but the reduced height simplifies the mesh and keeps the solution time reasonable. The dimensions below describe the complete motor, so extensive results must be scaled before extrapolation to the full motor.

The propellant is not included in the geometry. Its effect is represented by a prescribed gas temperature and convection coefficient on the inner surface of `phe0`. It should also be noted that ammonium perchlorate (AP) is the *oxidizer* in a composite propellant, not a fuel by itself. The composition of the fuel binder, any aluminum fraction, chamber pressure, and combustion history are not documented in the present model. Consequently, a wall heat flux cannot be derived from first principles: the current hot-side boundary condition remains an input that must be calibrated against a motor firing or supplied by a separate internal-flow model.

1.2 Dimensions Extracted from the 3D Model

The AP214 STEP file uses meters. Its cylindrical surfaces define the interface radii listed in Table 1. The model is a straight cylinder of length $L = 1.8288\,\mathrm{m}$, exactly 72 in; its bore diameter is 266.7 mm (10.5 in), and its outside diameter is 304.8 mm (12 in). The layers have a combined thickness of 19.05 mm.

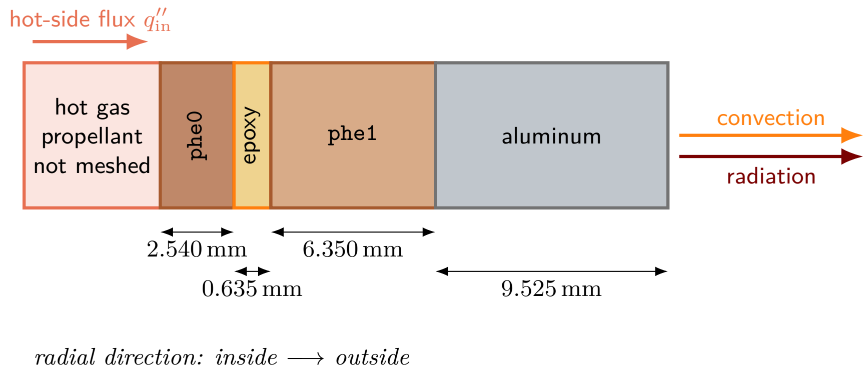

Figure 1. Radial layup extracted from the 3D model. Layer widths are schematic; the stated dimensions are exact.

Body	Inner radius (mm)	Outer radius (mm)	Thickness (mm)	Estimated initial mass (kg)
`phe0`	133.350	135.890	2.540	4.911
`epo`	135.890	136.525	0.635	1.143
`phe1`	136.525	142.875	6.350	12.742
`alu`	142.875	152.400	9.525	43.629

Table 1. Motor geometry. Masses are calculated using the model's initial densities and exclude end closures, seals, interfaces, and propellant.

The hot-side cylindrical area is $2\pi r_i L = 1.5323\,\mathrm{m}^2$, and the outside area is $2\pi r_o L = 1.7512\,\mathrm{m}^2$. These values will be useful for solver-independent energy and mass balances.

1.3 Materials and Interfaces

The inner body `phe0` uses the user material `PHENOLIC_PYROLYSIS`; `phe1` uses `PHENOLIC_VIRGIN`; the epoxy and aluminum bodies use the project's generic materials. All three concentric interfaces are program-controlled bonded thermal contacts. This configuration represents perfect adhesion with neither debonding nor thermal contact resistance. It is appropriate for an initial study, but it does not investigate delamination, epoxy charring, or the formation of a gas-filled gap.

The initial density is $1250\,\mathrm{kg\,m^{-3}}$ for virgin phenolic, $1150\,\mathrm{kg\,m^{-3}}$ for epoxy, and $2700\,\mathrm{kg\,m^{-3}}$ for aluminum. The aluminum alloy and exact epoxy grade are unspecified. Their associated properties are therefore generic approximations, not qualified material data.

2 Simulation 1: Volumetric Pyrolysis of the First Phenolic Layer

2.1 Theory and Derivation

2.1.1 Local Conservation of Energy

Consider a fixed material volume V of phenolic, bounded by ∂V, with no macroscopic motion of the solid. The first law of thermodynamics requires the rate of change of internal energy to equal the net heat input by conduction plus the volumetric heat generation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho u \,\mathrm{d}V = -\int_{\partial V} \boldsymbol{q}\cdot\boldsymbol{n}\,\mathrm{d}S + \int_V \rho\dot{q}_m\,\mathrm{d}V. \tag{1}$$

Here, u is the specific internal energy, ρ is the density of the condensed medium, $\dot{q}_m$ is a specific heat source, and $\boldsymbol{q}$ is the heat flux. Applying the divergence theorem to the surface integral and noting that the volume is arbitrary gives

$$\frac{\partial(\rho u)}{\partial t} = -\boldsymbol{\nabla}\cdot\boldsymbol{q} + \rho\dot{q}_m. \tag{2}$$

Using Fourier's law, $\boldsymbol{q} = -k\boldsymbol{\nabla}T$, and identifying $\partial u/\partial T$ with c_p in the Mechanical thermal formulation yields the form used in the model:

$$\rho c_p\frac{\partial T}{\partial t} = \boldsymbol{\nabla}\cdot(k\boldsymbol{\nabla}T) + \rho\dot{q}_m. \tag{3}$$

Pyrolysis is endothermic. If $\Delta H_p > 0$ is the enthalpy absorbed per kilogram of converted material and α is the degree of conversion, the specific heat source is

$$\dot{q}_m = -\Delta H_p\dot{\alpha}, \tag{4}$$

and therefore

$$\boxed{\rho c_p\frac{\partial T}{\partial t} = \boldsymbol{\nabla}\cdot(k\boldsymbol{\nabla}T) - \rho\Delta H_p\dot{\alpha}.} \tag{5}$$

This first simulation does not transport the enthalpy of the pyrolysis gases and does not move the hot boundary. Equation (5) is therefore a conduction–reaction equation, not a complete ablation model.

2.1.2 Conversion Variable and Mass Loss

At a given temperature, the model interpolates between the virgin state v and the char residue c. The apparent density of the solid is written as

$$\rho_s(T,\alpha) = (1-\alpha)\rho_v(T) + \alpha\rho_c(T). \tag{6}$$

The equivalent definition of conversion is

$$\alpha = \frac{\rho_v - \rho_s}{\rho_v - \rho_c}, \qquad 0 \le \alpha \le 1. \tag{7}$$

Thus, $\alpha = 0$ denotes virgin material and $\alpha = 1$ denotes the residual *char*. This distinction is essential: $\alpha = 1$ means neither a void nor a fully consumed element. In this single-reaction model, the rate of volatile matter released per unit volume is

$$\dot{\omega}_g''' = (\rho_v - \rho_c)\dot{\alpha}. \tag{8}$$

The other properties are mixed linearly:

$$X(T,\alpha) = (1-\alpha)X_v(T) + \alpha X_c(T), \qquad X \in \{k, c_p, \rho\}. \tag{9}$$

This rule is simple and monotonic, but it is not a universal constitutive law for porous composites. Its validation requires measurements at intermediate conversion states or a sensitivity analysis.

2.1.3 Arrhenius Kinetics and Integration over an Increment

The frequency of activated molecular events is represented by the Arrhenius rate constant

$$k_r(T) = A\exp\left(-\frac{E_a}{RT}\right), \tag{10}$$

where A is the pre-exponential factor, E_a is the activation energy, and $R = 8.314\,462\,618\,\mathrm{J\,mol^{-1}\,K^{-1}}$. Assuming that the remaining quantity of reactive sites is proportional to $(1-\alpha)^n$,

$$\boxed{\dot{\alpha} = k_r(T)(1-\alpha)^n.} \tag{11}$$

During the current increment $[t, t+\Delta t]$, the code evaluates T at the midpoint temperature and treats k_r as constant. Let $\beta = 1-\alpha$. Then $\dot{\beta} = -k_r\beta^n$. For $n = 1$,

$$\int_{\beta_0}^{\beta_1} \frac{\mathrm{d}\beta}{\beta} = -k_r \int_t^{t+\Delta t} \mathrm{d}t, \tag{12}$$

hence

$$\beta_1 = \beta_0 \exp(-k_r\Delta t), \qquad \boxed{\alpha_1 = 1-(1-\alpha_0)\exp(-k_r\Delta t).} \tag{13}$$

For $n \neq 1$, integration gives

$$\frac{\beta_1^{1-n} - \beta_0^{1-n}}{1-n} = -k_r\Delta t, \tag{14}$$

or

$$\boxed{\beta_1 = \left[\beta_0^{1-n} - (1-n)k_r\Delta t\right]^{1/(1-n)}, \qquad \alpha_1 = 1-\beta_1.} \tag{15}$$

The bracketed term is bounded to prevent a nonphysical value. The stored rate is the average over the increment, $\dot{\alpha} = (\alpha_1 - \alpha_0)/\Delta t$. The code requests a time-step cutback if $\Delta\alpha > 0.05$, thereby limiting excessively abrupt numerical conversion. The *current increment* specifically means the substep attempted or accepted by the solver between two converged time points, not the complete seven-second load step.

2.1.4 Finite-Element Weak Form

Multiplying Equation (5) by a test function N_i and integrating the conduction term by parts gives

$$\int_V N_i \rho c_p \frac{\partial T}{\partial t}\,\mathrm{d}V + \int_V \boldsymbol{\nabla} N_i \cdot k \boldsymbol{\nabla} T\,\mathrm{d}V = -\int_{\partial V} N_i \boldsymbol{q}\cdot\boldsymbol{n}\,\mathrm{d}S \tag{16}$$

$$-\int_V N_i \rho \Delta H_p \dot{\alpha}\,\mathrm{d}V. \tag{17}$$

After applying the approximation $T = \sum_j N_j T_j$, the matrix form is

$$\boldsymbol{C}(T,\alpha)\dot{\boldsymbol{T}} + \boldsymbol{K}(T,\alpha)\boldsymbol{T} = \boldsymbol{f}_q - \boldsymbol{f}_p(T,\alpha). \tag{18}$$

The dependence of k, c_p, ρ, and $\dot{\alpha}$ on temperature and history makes the problem nonlinear. `UserMatTh` is called at every integration point and returns the heat flux, its tangent, the heat capacity, the density, the specific heat source, and the state variables [1].

2.2 Current ANSYS Mechanical Configuration

2.2.1 Discretization, Contacts, and Boundary Conditions

All four bodies use `SOLID279` three-dimensional thermal elements. The inspected export contains approximately 482 900 solid elements and 2 242 327 equations. Their distribution is given in Table 2. The eight layers through `phe0` yield an average radial thickness of only 0.318 mm, which is suitable for resolving a pyrolysis front. Even with the axial truncation described above, this radial resolution remains computationally expensive.

Body	Elements	Mean radial layers	Material
`phe0`	315536	8	`PHENOLIC_PYROLYSIS`
`epo`	9516	1	generic epoxy
`phe1`	59028	6	`PHENOLIC_VIRGIN`
`alu`	98820	9	generic aluminum

Table 2. Mesh of the axially truncated domain exported by Mechanical.

The three contact pairs are bonded: `phe0_ext-epo_int`, `epo_ext-phe1_int`, and `phe1_ext-alu_int`. The initial temperature is $T_0 = 295.15\,\mathrm{K}$. The inner face is subjected to

$$q''_{\mathrm{in}} = h_{\mathrm{in}}(T_g - T_s), \qquad h_{\mathrm{in}} = 1000\,\mathrm{W\,m^{-2}\,K^{-1}}, \quad T_g = 2504\,\mathrm{K}. \tag{19}$$

This corresponds to 2230.85 °C in the Mechanical interface. On the outside, the model combines natural convection with $h_{\mathrm{out}} = 8\,\mathrm{W\,m^{-2}\,K^{-1}}$ toward 295.15 K and gray-body radiation:

$$q''_{\mathrm{rad}} = \varepsilon\sigma(T_s^4 - T_\infty^4), \qquad \varepsilon_{\mathrm{Al}} = 0.25. \tag{20}$$

The coefficients h_{in} and h_{out}, the gas temperature, and the emissivity are project assumptions, not outputs from a combustion calculation.

The study is a full 7 s transient thermal analysis with automatic time stepping: $\Delta t_{\text{initial}} = 0.05\,\text{s}$, $\Delta t_{\text{min}} = 0.001\,\text{s}$, $\Delta t_{\text{max}} = 0.05\,\text{s}$. The user routine adds its own criterion, $\Delta\alpha \leq 0.05$. State-variable results are requested at every substep through `OUTRES,SVAR,ALL`. The inspected configuration launches eight DMP processes; the routine uses no mutable global variables and is therefore also suitable for a future distributed run, provided that the DLL is accessible to every process.

2.2.2 Thermal Properties and Kinetic Parameters

Table 3 reproduces the property curves actually passed to `UserMatTh`. Mechanical uses degrees Celsius in this export, whereas the table is presented in kelvins. ANSYS interpolates the properties between the tabulated points; extrapolation beyond the final point should be avoided.

T (K)	k_v (W m^{-1} K^{-1})	k_c	$c_{p,v}$ (J kg^{-1} K^{-1})	$c_{p,c}$	ρ_v (kg m^{-3})	ρ_c
300	0.25	0.30	900	800	1250	700
600	0.32	0.35	1100	900	1200	650
900	0.36	0.40	1300	1000	1150	600
1050	0.36	0.44	1300	1250	1150	600
1200	0.36	0.49	1300	1550	1150	600
1500	0.36	0.68	1300	1950	1150	600
1700	0.36	0.81	1300	2060	1150	600
2200	0.36	1.24	1300	2085	1150	600
2750	0.36	1.73	1300	2090	1150	600

Table 3. Currently encoded virgin/char property curves for the `PHENOLIC_PYROLYSIS` material. These are project engineering curves, not yet a certified characterization of the manufactured grade.

Parameter	**Current value**	**Status and source**
A	333 s^{-1}	Value read from `phenolic_pyrolysis_native.apdl`; its experimental origin is not included in the project. To be recalibrated using TGA at multiple heating rates.
E_a	$64.081\,\text{kJ}\,\text{mol}^{-1}$	Same status as A; the pair (A, E_a) must be identified jointly.
n	1	Assumption of a global first-order reaction.
ΔH_p	$418\,\text{kJ}\,\text{kg}^{-1}$	Endothermic enthalpy encoded in the project; to be replaced by a DSC measurement of the actual resin/fiber system.
R	$8.314\,46\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$	Universal gas constant.
$\Delta\alpha_{\text{max}}$	0.05 per increment	Numerical criterion encoded in `usermatth.F`, not a physical property.

Table 4. Pyrolysis kinetics used in Simulation 1.

The virgin phenolic body `phe1` uses the virgin columns of Table 3 without conversion. The generic epoxy has $\rho = 1150\,\text{kg}\,\text{m}^{-3}$; its conductivity is $0.25\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$ and $0.30\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$ at 300 K and 600 K, respectively, and its specific heat capacity is $1000\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$ and $1100\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$ at the same temperatures. The generic aluminum has $\rho = 2700\,\text{kg}\,\text{m}^{-3}$, $k = 167\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$, and $c_p = 896\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$. Within the project, neither material has a complete property curve extending to 2750 K; constant extrapolation is a major risk if heat reaches the outer layers. A qualified study requires the epoxy grade and aluminum alloy to be identified, followed by the addition of their temperature-dependent properties, transitions, decomposition, or melting over the temperature range actually reached.

The char trends—increasing conductivity at high temperature, mass loss, and high heat capacity—are qualitatively consistent with historical charring ablator models [7, 4]. They do not, however, justify the specific numerical values for an unidentified phenolic. NASA uses TGA to measure mass loss and char yield, DSC to determine c_p and enthalpies, and LFA to determine thermal diffusivity/conductivity [9].

2.2.3 MAPDL Code

The following MAPDL block is the file actually associated with the inner material. It creates nine `TB,USER` points, eleven properties at each temperature, and two state variables: `SVAR1`=α and `SVAR2`=$\dot{\alpha}$. The `matid` variable is resolved by the Mechanical command context; it must not be hard-coded without first checking the identifier produced by the export.

Complete MAPDL listing — file `phenolic_pyrolysis_native.apdl`.

```
! ------------------------------------------------------------
! Prebuilt native Fortran UserMatTh material for inner body phe0
! Solver units: m, kg, s, W, degree C
! ------------------------------------------------------------

/COM,Prebuilt Fortran UserMatTh pyrolysis model requested

PYRA  = 333.0
PYREA = 64081.0
PYRN  = 1.0
PYRDH = 418000.0
TO_K  = 273.15

! TB,USER properties:
! 1  virgin conductivity
! 2  char conductivity
! 3  virgin specific heat
! 4  char specific heat
! 5  virgin density
! 6  char density
! 7  Arrhenius A
! 8  activation energy Ea
! 9  reaction order n
! 10 positive endothermic pyrolysis enthalpy
! 11 Celsius-to-Kelvin offset

TB,USER,matid,9,11,THERM

TBTEMP,26.85
TBDATA,1,0.25,0.30,900,800,1250,700
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,326.85
TBDATA,1,0.32,0.35,1100,900,1200,650
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,626.85
TBDATA,1,0.36,0.40,1300,1000,1150,600
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,776.85
TBDATA,1,0.36,0.44,1300,1250,1150,600
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,926.85
TBDATA,1,0.36,0.49,1300,1550,1150,600
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,1226.85
TBDATA,1,0.36,0.68,1300,1950,1150,600
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,1426.85
TBDATA,1,0.36,0.81,1300,2060,1150,600
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,1926.85
TBDATA,1,0.36,1.24,1300,2085,1150,600
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

TBTEMP,2476.85
TBDATA,1,0.36,1.73,1300,2090,1150,600
TBDATA,7,PYRA,PYREA,PYRN,PYRDH,TO_K

! SVAR1 = alpha; SVAR2 = alpha rate
TB,STATE,matid,,2
TBDATA,1,0.0,0.0
```

2.2.4 Fortran `UserMatTh` Code

The native routine below is the one compiled into `usermatthLib.dll`. It interpolates the `TB,USER` properties, integrates the kinetics analytically, updates both state variables, and returns $\boldsymbol{q} = -k\boldsymbol{\nabla}T$, $\partial u/\partial T = c_p$, the density, and the specific heat generation $-\Delta H_p \dot{\alpha}$. According to the ANSYS documentation, `hgen` is indeed a source per unit mass for purely thermal elements [1].

Complete Fortran listing — file `usermatth.F`.

```fortran
*deck,usermatth                                USERDISTRIB
      subroutine usermatth(matId, elemId, kDomIntPt, kLayer,
     &                     kSectPt, ldstep, isubst, keycut,
     &                     ncomp, nStatev, nProp, Time, dTime,
     &                     Temp, dTemp, tgrad, ustatev, prop,
     &                     coords, dudt, dudg, flux, dfdt,
     &                     dfdg, cutFactor, hgen, dens,
     &                     var1, var2, var3, var4, var5, var6)
c
c     Native UserMatTh model for phenolic pyrolysis.
c
c     Properties in TB,USER (SI units):
c       1  virgin thermal conductivity       [W/(m K)]
c       2  char thermal conductivity          [W/(m K)]
c       3  virgin specific heat               [J/(kg K)]
c       4  char specific heat                  [J/(kg K)]
c       5  virgin density                      [kg/m3]
c       6  char density                        [kg/m3]
c       7  Arrhenius pre-exponential factor A  [1/s]
c       8  activation energy Ea                [J/mol]
c       9  reaction order n                    [-]
c      10  positive endothermic enthalpy       [J/kg]
c      11  temperature offset                  [K]
c
c     State variables:
c       1  alpha, 0 = virgin and 1 = char
c       2  alpha rate                          [1/s]
c
c     Kinetics:
c       d(alpha)/dt = A exp(-Ea/(R T)) (1-alpha)**n
c
c     The equation is integrated analytically over the current time
c     increment.  No mutable global state is used, so the routine is
c     suitable for shared- and distributed-memory solves.
c
      implicit none

      integer matId, elemId, kDomIntPt, kLayer, kSectPt
      integer ldstep, isubst, keycut, ncomp, nStatev, nProp
      integer i, j

      double precision Time, dTime, Temp, dTemp
      double precision tgrad(ncomp), ustatev(nStatev)
      double precision prop(nProp), coords(3)
      double precision dudt, dudg(ncomp), flux(ncomp)
      double precision dfdt(ncomp), dfdg(ncomp,ncomp)
      double precision cutFactor, hgen, dens
      double precision var1, var2, var3, var4, var5, var6

      double precision gasr, minTemp, maxDAlpha, minCut
      parameter (gasr=8.31446261815324d0)
      parameter (minTemp=1.0d0)
      parameter (maxDAlpha=5.0d-2)
      parameter (minCut=1.0d-1)

      double precision kv, kc, cpv, cpc, rhov, rhoc
      double precision apre, ea, ordern, deltaH, tempOffset
      double precision temperatureMid, exponent, rateConstant
      double precision alphaOld, alphaNew, alphaDot, deltaAlpha
      double precision betaOld, betaNew, power, bracket
      double precision conductivity, specificHeat, densityNew

c     Always initialize every output controlled by UserMatTh.
      keycut = 0
      cutFactor = 1.0d0
      dudt = 1.0d0
      do i = 1, ncomp
         dudg(i) = 0.0d0
         flux(i) = 0.0d0
         dfdt(i) = 0.0d0
         do j = 1, ncomp
            dfdg(i,j) = 0.0d0
         enddo
      enddo

```

```
c     A cutback is safer than accessing an incomplete material table.
      if (nProp .lt. 11 .or. nStatev .lt. 2) then
         keycut = 1
         cutFactor = minCut
         return
      endif

      kv         = prop(1)
      kc         = prop(2)
      cpv        = prop(3)
      cpc        = prop(4)
      rhov       = prop(5)
      rhoc       = prop(6)
      apre       = prop(7)
      ea         = prop(8)
      ordern     = prop(9)
      deltaH     = prop(10)
      tempOffset = prop(11)

c     Reject nonphysical values without allowing NaNs into the solve.
      if (kv .le. 0.0d0 .or. kc .le. 0.0d0 .or.
     &    cpv .le. 0.0d0 .or. cpc .le. 0.0d0 .or.
     &    rhov .le. 0.0d0 .or. rhoc .le. 0.0d0 .or.
     &    apre .lt. 0.0d0 .or. ea .lt. 0.0d0 .or.
     &    ordern .le. 0.0d0 .or. deltaH .lt. 0.0d0) then
         keycut = 1
         cutFactor = minCut
         return
      endif

      alphaOld = dmax1(0.0d0, dmin1(1.0d0, ustatev(1)))
      temperatureMid = Temp + 0.5d0*dTemp + tempOffset
      temperatureMid = dmax1(temperatureMid, minTemp)

      exponent = -ea/(gasr*temperatureMid)
      exponent = dmax1(-700.0d0, dmin1(700.0d0, exponent))
      rateConstant = apre*dexp(exponent)

      alphaNew = alphaOld
      if (dTime .gt. 0.0d0 .and. rateConstant .gt. 0.0d0
     &    .and. alphaOld .lt. 1.0d0) then
         betaOld = dmax1(1.0d0-alphaOld, 0.0d0)
         if (dabs(ordern-1.0d0) .le. 1.0d-10) then
            betaNew = betaOld*dexp(-rateConstant*dTime)
         else
            power = 1.0d0-ordern
            bracket = betaOld**power
     &                - power*rateConstant*dTime
            if (bracket .le. 0.0d0) then
               betaNew = 0.0d0
            else
               betaNew = bracket**(1.0d0/power)
            endif
         endif
         alphaNew = 1.0d0-betaNew
         alphaNew = dmax1(alphaOld, dmin1(1.0d0, alphaNew))
      endif

      deltaAlpha = alphaNew-alphaOld
      alphaDot = 0.0d0
      if (dTime .gt. 0.0d0) alphaDot = deltaAlpha/dTime

c     Linear virgin/char mixture at the updated conversion.
      conductivity = (1.0d0-alphaNew)*kv + alphaNew*kc
      specificHeat = (1.0d0-alphaNew)*cpv + alphaNew*cpc
      densityNew   = (1.0d0-alphaNew)*rhov + alphaNew*rhoc

      dudt = specificHeat
      dens = densityNew
      do i = 1, ncomp
         flux(i) = -conductivity*tgrad(i)
         dfdg(i,i) = -conductivity
      enddo

      ustatev(1) = alphaNew
      ustatev(2) = alphaDot

c     For pure thermal elements hgen is heat generation per unit mass.
c     Positive deltaH is endothermic, hence the minus sign.
      hgen = hgen-deltaH*alphaDot

c     Limit conversion to five percent per accepted increment.
      if (deltaAlpha .gt. maxDAlpha) then
         keycut = 1
         cutFactor = maxDAlpha/dmax1(deltaAlpha, 1.0d-30)
```

```
         cutFactor = dmax1(minCut, dmin1(0.95d0, cutFactor))
      endif

      return
      end
```

The Windows x64 build uses Visual Studio 2022 C++ and Intel oneAPI Classic Fortran 2023.1. The `build_usermatth_x64.cmd` script compiles with `/O2 /MD`, links the ANSYS v261 customization libraries, and verifies that the `USERMATTH` symbol is exported. This toolchain must remain aligned with the solver version.

2.2.5 Step-by-Step Simulation Preparation

1. **Identify the physical materials.** Record the manufacturer, grade, reinforcement fraction and orientation of the phenolic, the epoxy system, and the aluminum alloy. In the absence of this information, retain an explicit "preliminary data" designation.
2. **Characterize the phenolic.** Perform TGA in an inert atmosphere at several heating rates to obtain $\rho(T)$, the char yield, and the kinetics; use DSC for $c_p(T)$ and ΔH_p, LFA for $k(T)$, and then measure the apparent density of the char. First fit A, E_a, n against all TGA curves and validate against a heating rate not used in the fit. A multiple-reaction kinetic model is preferable if a single law cannot reproduce the peaks.
3. **Complete Engineering Data.** Enter the virgin and char tables up to at least the maximum gas temperature, with explicit units. Add the epoxy and aluminum curves over the temperature range actually reached. Do not blindly extrapolate a polymer to 2750 K.
4. **Assign the bodies.** In Mechanical, verify `phe0`=`PHENOLIC_PYROLYSIS`, `phe1`=`PHENOLIC_VIRGIN`, `epo`=epoxy, and `alu`=aluminum. Check the assignment in the `ds.dat` export, not solely from the interface color.
5. **Prepare named selections.** Retain the four bodies, the three interface pairs, the hot inner face, and the outer face. Stable names prevent MAPDL commands from being attached to a fragile geometric selection.
6. **Define the contacts.** For this simulation, use bonded thermal contacts and verify that each pair is closed. If an adhesive is meshed as a volume, do not also add a contact resistance that would count the same effect twice.
7. **Mesh and test mesh independence.** Retain at least eight layers through `phe0` for the reference calculation, then compare the wall temperature and the $\alpha = 0.5$ depth against a finer radial mesh. Mesh independence must be assessed using these observables, not only the maximum temperature.
8. **Build the DLL.** From a properly initialized x64 command prompt, run `build_usermatth_x64.cmd`; check the return code, `compile.log`, `link.log`, and the `USERMATTH` export. Copy the DLL to the solver directory, or otherwise make it accessible there, before starting the run.
9. **Insert the MAPDL block.** Under the thermal-analysis branch, insert a command object that applies the contents of Section 2.2.3 to the `phe0` material. After generating the solver input, verify the presence of `TB,USER`, `TB,STATE`, and nine `TBTEMP` entries.
10. **Define the loading.** Prescribe 295.15 K initially, the hot convection of Equation (19), and then the outer convection and radiation. Ultimately, replace $T_g(t)$ and $h(t)$ with profiles from a test or a chamber calculation. Also vary h over at least 500 to 2000 $\mathrm{W\,m^{-2}\,K^{-1}}$ to quantify its influence.
11. **Set time stepping and convergence controls.** Use automatic time stepping between 0.001 s and 0.05 s, retain the $\Delta\alpha \leq 0.05$ limit, and save `SVAR`. A single 7 s step would eliminate resolution of the front and would not be physically acceptable, even though the local law is integrated analytically.
12. **Verify solver startup.** In the solver output, confirm that the user library is loaded, that the requested number of processes is used, that no `UserMatTh` error occurs, and that the state variables are present. First run a coupon or a very small sector for 0.1 s.
13. **Validate using conservation balances.** For every result set, check $0 \leq \alpha \leq 1$, the monotonicity of α in time, the heat balance, and the integrated volatile mass obtained from Equation (8). Finally, compare a case without pyrolysis and a simplified adiabatic case against an analytical conduction solution.

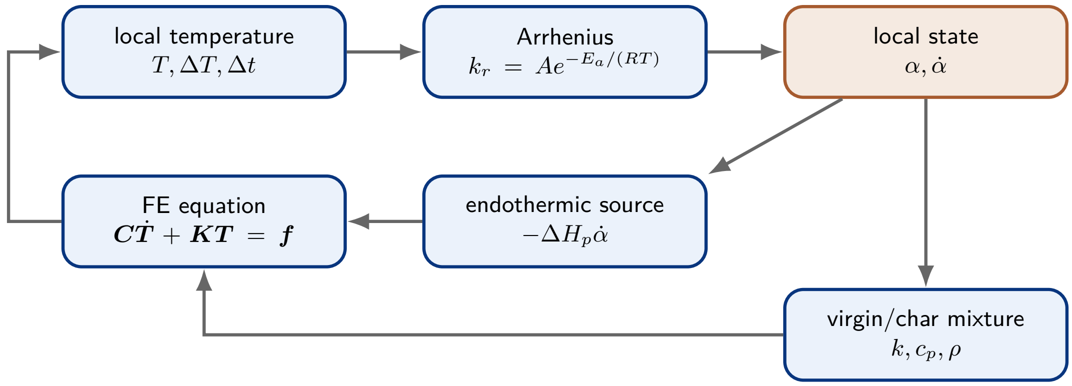

Figure 2. Local coupling in Simulation 1. Conversion modifies both the heat source and the properties at the following substep.

2.3 Simulation 1 Results

2.3.1 Data Scope and Numerical Convergence

The analyzed results come from the eight DMP partitions `file0.rth`–`file7.rth`, the `file.nlh` history, and the `solve.out` output [12]. The last saved and readable state is set 140 at

$$t = 6.385\,616\,94\,\mathrm{s}.$$

The requested solution was intended to reach 7 s; the values below therefore describe a converged intermediate state rather than the final seven-second result. Any linear extrapolation to 7 s is not a substitute for the missing 0.614 s of computation.

At this time, the solver has accepted 140 substeps and accumulated 293 equilibrium iterations. The last substep is 0.05 s, converges in two iterations, and ends with a heat-flow norm of 0.01694, well below the 0.6255 criterion. The reported wall time is approximately 90 883 seconds, or 25.2 h, on eight DMP processes. These indicators show good *numerical* convergence of the saved state; they are neither a validation of the material properties nor evidence of mesh independence.

2.3.2 Temperature Field

Table 5 collects the robust extrema obtained by scanning the partitions. Interface nodes can be duplicated by contact elements; partition-wise nodal averages are therefore not used as volume averages.

Observable at $t = 6.3856\,\mathrm{s}$	**Temperature (°C)**	**Temperature (K)**
Hot inner face, nodal range	1754.06–1755.39	2027.21–2028.54
Maximum in `phe0`	1755.39	2028.54
Maximum in the epoxy	453.41	726.56
Maximum in `phe1`	208.05	481.20
Maximum in the aluminum	22.0012	295.1512
Maximum on the outer aluminum face	22.000005	295.150005

Table 5. Temperatures extracted from the last converged state.

The hot face is nearly isothermal and reaches 1755.39 °C, or 2028.54 K. It remains approximately 475.5 K below the imposed gas temperature. The prescribed convection therefore corresponds at this instant to an inward heat flux of only $0.475 - -0.477\,\mathrm{MW\,m^{-2}}$, compared with approximately $2.21\,\mathrm{MW\,m^{-2}}$ immediately after the hot condition is applied as a step. This flux is a direct consequence of h_{in} and T_g, not an independent prediction of combustion.

The most concerning design point is not yet the aluminum, which remains essentially at its initial temperature, but the epoxy: its maximum of 453.41 °C is 726.56 K, beyond the final material property point currently defined at 600 K. Its temperature is therefore computed using constant extrapolation of a generic material, while decomposition and loss of adhesion are not represented. The maximum in `phe1`, 208.05 °C, shows that the thermal wave has crossed the epoxy but has not traversed the 6.35 mm second phenolic layer. The nearly ambient aluminum temperature must not be extrapolated to the head-end, nozzle, or attachment regions absent from the 50 mm slice, nor to the heat-soak phase after motor shutdown.

2.3.3 Pyrolysis Conversion `SVAR1`

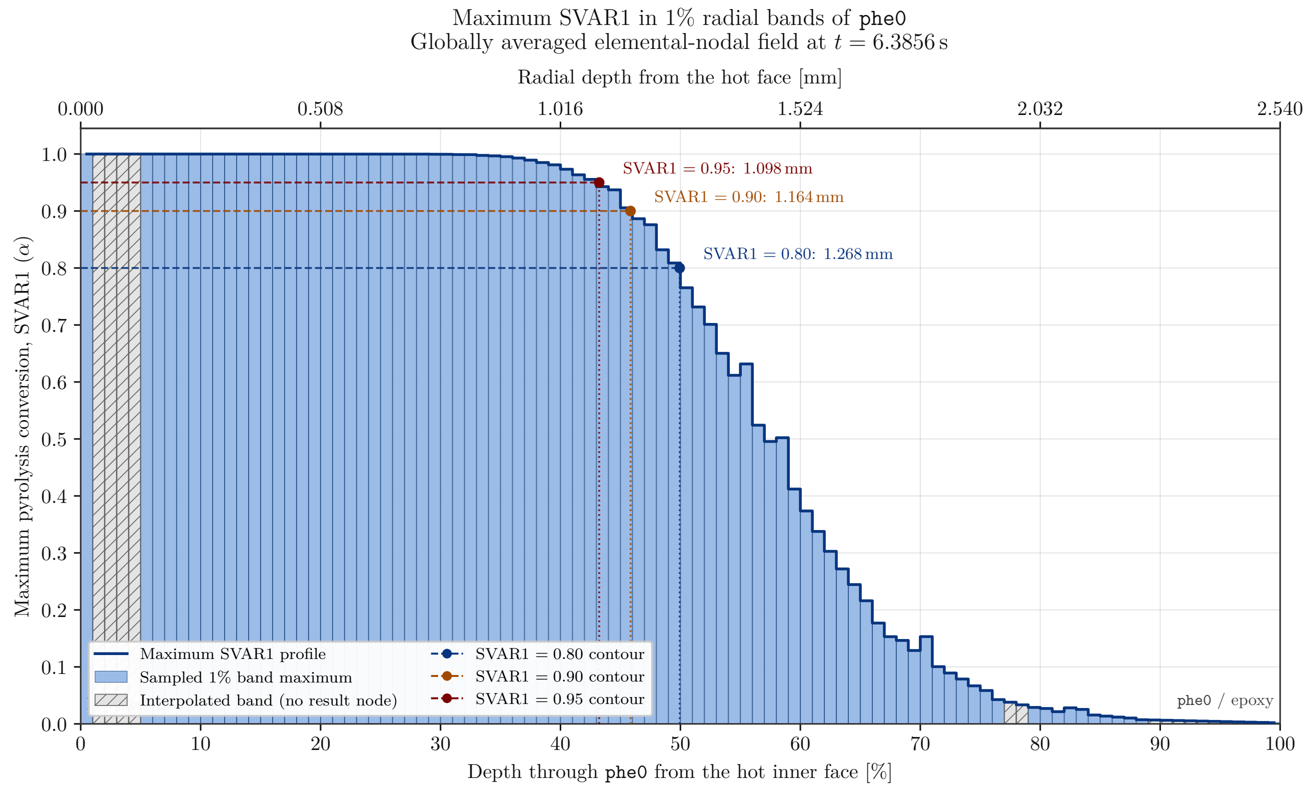

Figure 3. Maximum `SVAR1` in one-percent radial bands of `phe0` at the last converged state. Hatched bands contain no result node and are filled by linear interpolation.

Figure 3 shows the maximum radial envelope of the globally averaged elemental-nodal field. The global maximum reaches $\alpha = 1$: the zone near the hot face is fully converted according to the kinetic law. This does not mean that the char has been removed or that a void has been created.

The isoconversion depths are given in Table 6. The region with at least 95% conversion reaches 1.098 mm, or 43.2% of the `phe0` thickness; the 80% level reaches almost exactly half of the layer.

Threshold α_c	**Depth** d_{α_c} **(mm)**	**Fraction of** `phe0`	**Apparent speed** d_{α_c}/t (mm s^{-1})
0.80	1.26845	49.94%	0.19864
0.90	1.16440	45.84%	0.18235
0.95	1.09800	43.23%	0.17195

Table 6. Depth and apparent mean speed of the conversion fronts.

The speeds in the last column are only the ratios d_{α_c}/t. They assume that the contour starts at the hot face at $t = 0$ and provide a mean penetration speed since ignition. They are not instantaneous front speeds. A rigorous front speed in mm s^{-1} must be calculated by differencing the positions of the same contour at several saved times.

One band in the figure is 25.4 µm wide, whereas the mean radial thickness of one `phe0` element is 0.318 mm. The 100 bars improve profile readability and integration, but do not create a physical resolution of 100 cells. Sixteen bands without a node had to be interpolated; the front-location uncertainty remains on the order of a fraction of an element and must be measured through mesh refinement.

2.3.4 Time Evolution of Conversion

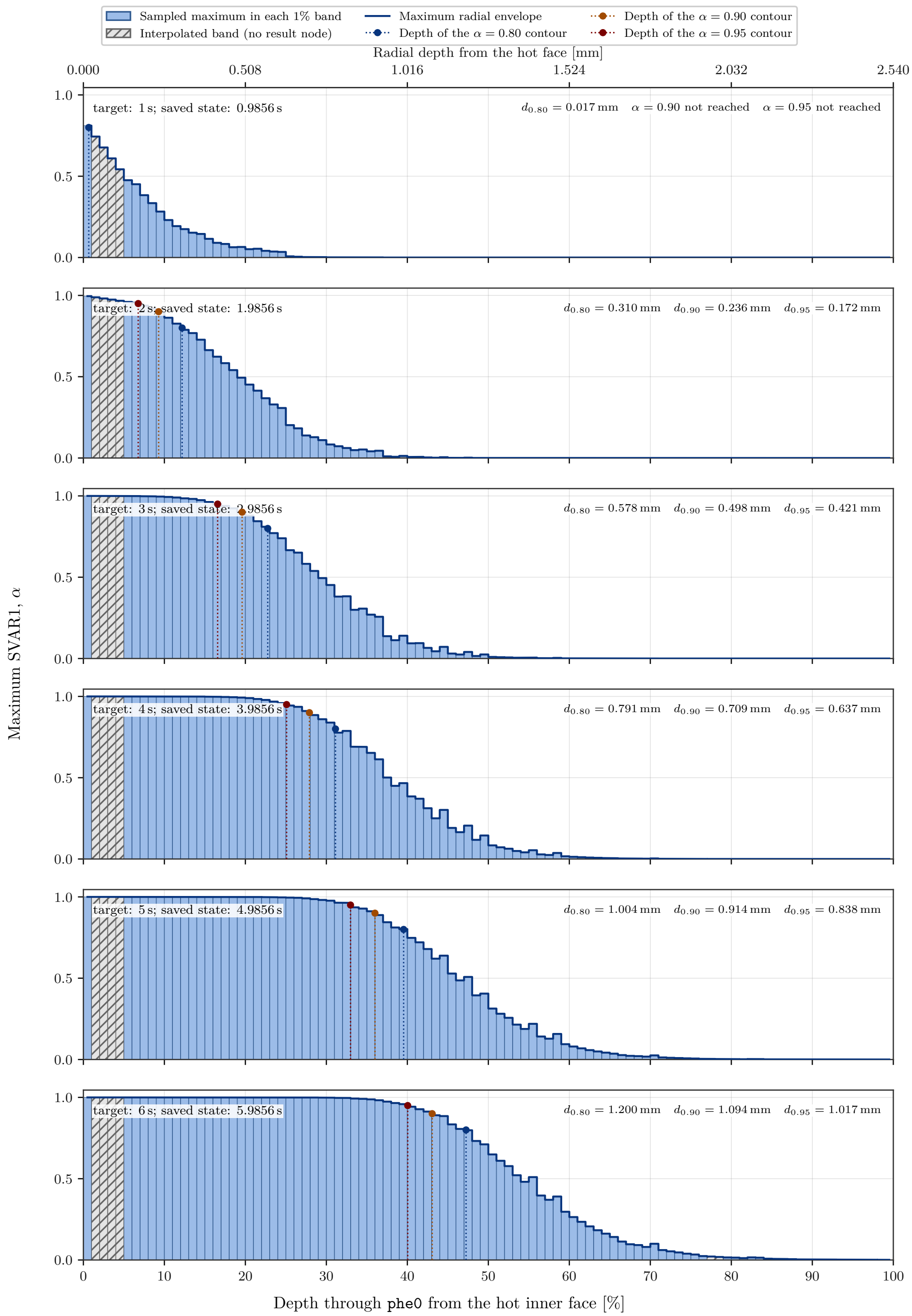

Figure 4. Evolution of the maximum radial `SVAR1` in `phe0` for the saved states closest to 1 through 6 s. The six plots are stacked vertically and use identical axes. The dotted lines and points mark the depths of the 0.80, 0.90, and 0.95 isoconversion levels when they are reached.

Figure 4 complements, without replacing, the last-converged-state profile in Figure 3. It applies exactly the same elemental-nodal averaging and the same one-percent radial bands to the six saved states closest to $t = 1, \ldots, 6$ s. The available outputs precede each integer second by 0.014 383 s; the time actually read is shown in every panel so that a temporal interpolation is not presented as an ANSYS result.

Table 7 collects the values used in the annotations. At 0.9856 s, the maximum value is still only $\alpha = 0.811$, and the 0.80 contour barely enters the layer: the 0.90 and 0.95 levels have not yet been reached. This is a thermal initiation phase, not an already established front.

Target time (s)	**Saved time (s)**	$\alpha_{\max}$	$d_{0.80}$ **(mm)**	$d_{0.90}$ **(mm)**	$d_{0.95}$ **(mm)**
1	0.9856	0.81114	0.01691	—	—
2	1.9856	0.99560	0.30979	0.23605	0.17200
3	2.9856	0.99997	0.57818	0.49786	0.42106
4	3.9856	1.00000	0.79069	0.70892	0.63735
5	4.9856	1.00000	1.00433	0.91446	0.83788
6	5.9856	1.00000	1.20013	1.09392	1.01716

Table 7. Time progression of the isoconversion depths computed from the maximum radial envelope.

Between the states near 1 and 2 s, the 0.80 contour advances at an incremental mean speed of 0.293 mm s^{-1}. This speed subsequently falls to 0.268 mm s^{-1}, then approximately 0.213 mm s^{-1}, 0.214 mm s^{-1}, and 0.196 mm s^{-1} over the successive intervals. Once it has appeared, the 0.95 contour slows in a similar manner, from 0.249 mm s^{-1} between 2 and 3 s to 0.179 mm s^{-1} between 5 and 6 s. The general trend is therefore a slowing front, with a small non-monotonic variation consistent with the substeps, nodal smoothing, and radial mesh resolution.

The profiles also show that there is no infinitely thin pyrolysis interface: a broad transition zone separates the almost fully converted char from material that is still virgin. Near 6 s, the 0.95 contour is at 1.017 mm, the 0.80 contour is at 1.200 mm, and measurable conversion extends substantially farther. At the last available state, 6.3856 s, these depths reach 1.098 mm and 1.268 mm, respectively, extending the trend of the six panels without a discontinuity.

For design, a single d/t speed therefore hides both the initiation delay and the subsequent slowing. Any extrapolation to the end of firing must state the chosen conversion threshold and use at least a recent incremental speed, rather than conflating conversion, char depth, and geometric recession. Finally, because every curve is a maximum radial envelope, it remains conservative for reconstructed conversion but represents neither a volume average nor a directly measured ablation depth.

2.3.5 Local Pyrolysis Rate `SVAR2`

`SVAR2` is a local conversion rate in s^{-1}, distinct from the geometric speed in Table 6. The global maximum over the partitions containing `phe0` is

$$\boxed{\dot{\alpha}_{\max} = 0.326\,660\,\mathrm{s}^{-1}.}$$

It occurs 1.442 21 mm from the hot face, or 56.78% through the thickness, where $\alpha = 0.54721$ and $T = 956.67\,^\circ\mathrm{C} = 1229.82\,\mathrm{K}$. This position is physically consistent: near the surface, the $1-\alpha$ factor is almost zero because the material is already converted; farther into the material, the temperature is too low to activate the Arrhenius law rapidly. The maximum rate therefore lies in the transition zone.

Over the last 0.05 s substep, this maximum corresponds to $\Delta\alpha \simeq 0.01633$, well below the numerical limit of 0.05. The `UserMatTh` criterion therefore does not force an additional reduction of the last increment. At this point,

$$k_r = \frac{\dot{\alpha}}{1-\alpha} = 0.7214\,\mathrm{s}^{-1}, \qquad \tau_r = k_r^{-1} = 1.386\,\mathrm{s}.$$

If the temperature were held artificially constant, it would take approximately 3.05 s to progress from $\alpha = 0.547$ to $\alpha = 0.95$. The local temperature is still changing in reality, so this characteristic time is not an extrapolation to the end of the firing.

With $\rho_v - \rho_c \simeq 550\,\mathrm{kg\,m^{-3}}$ at this temperature, the maximum local source suggested for Simulation 2 would be

$$\dot{\omega}'''_{g,\max} \simeq (\rho_v - \rho_c)\dot{\alpha}_{\max} = 180\,\mathrm{kg\,m^{-3}\,s^{-1}}.$$

The mass-specific thermal sink already applied by Simulation 1 is locally $-\Delta H_p \dot{\alpha} \simeq -136.5\,\mathrm{kW\,kg^{-1}}$. Gas generation is therefore potentially significant, but its transport, pressure, and heat exchange are not solved in the current model.

2.3.6 Time Evolution of the Local Rate

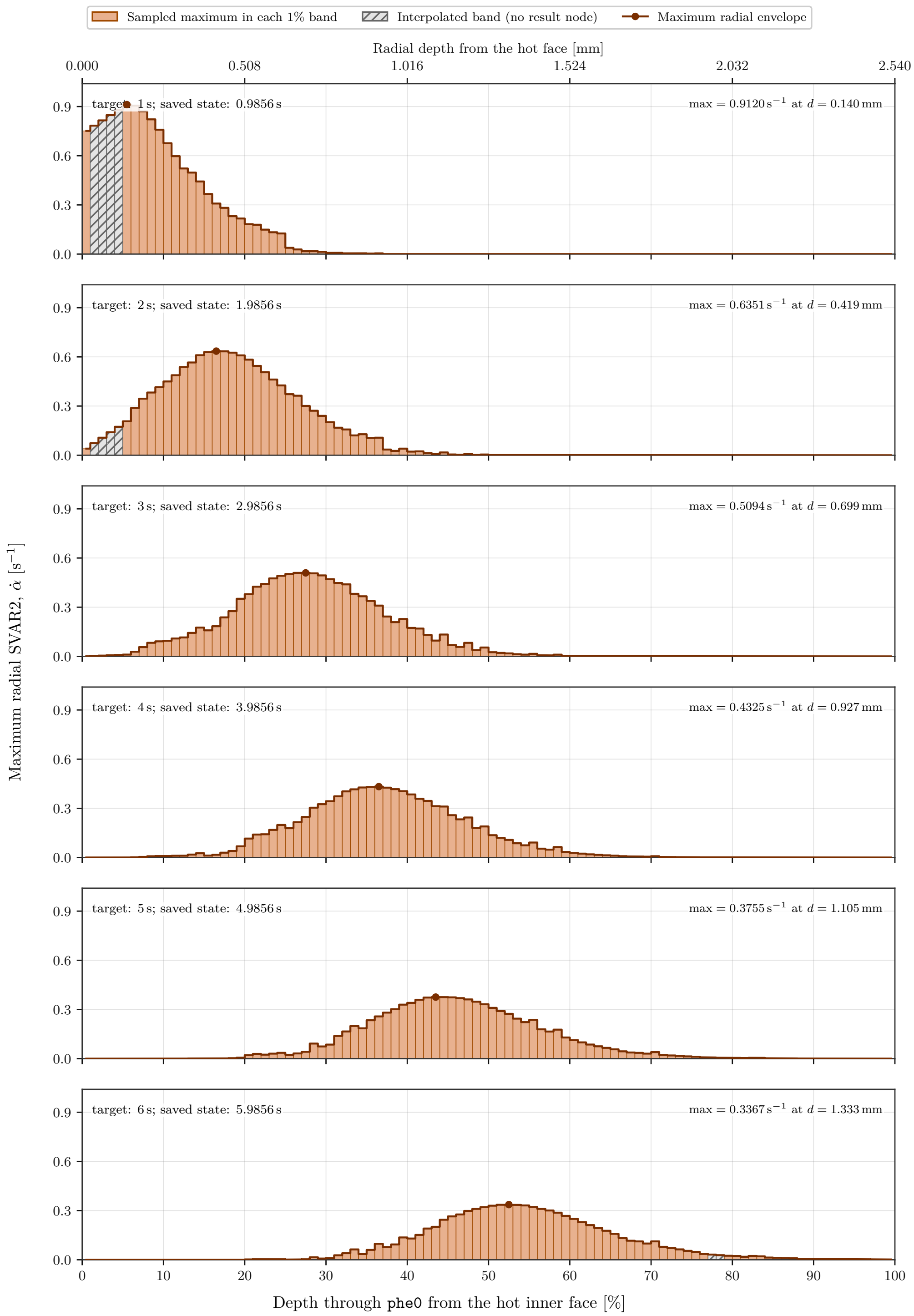

Figure 5. Evolution of the maximum radial `SVAR2` in `phe0`, for the saved states nearest 1 to 6 s. Every panel uses the same vertical scale; hatched bands contain no result node and are linearly interpolated.

Figure 5 applies to `SVAR2` the same time study as Figure 4. Table 8 shows both the decrease of the maximum rate and its motion inwards. Between the states near 1 and 6 s, the maximum drops from $0.9120\,\mathrm{s}^{-1}$ to $0.3367\,\mathrm{s}^{-1}$, a 63.1% decrease, while its depth increases from 0.140 mm to 1.333 mm.

Target time (s)	**Saved time (s)**	$\dot{\alpha}_{\max}$ (s^{-1})	**Depth of maximum (mm)**
1	0.9856	0.9120	0.140
2	1.9856	0.6351	0.419
3	2.9856	0.5094	0.699
4	3.9856	0.4325	0.927
5	4.9856	0.3755	1.105
6	5.9856	0.3367	1.333

Table 8. Time migration of the maximum radial pyrolysis rate.

The trajectory is not a geometric recession. It identifies the location where the product of the Arrhenius factor and the remaining virgin fraction is largest. The hot face saturates rapidly, so $1 - \alpha$ becomes nearly zero there; the maximum of $\dot{\alpha}$ then moves towards deeper material that is still partly virgin but sufficiently hot. The decrease of the maximum reflects thermal attenuation with depth, not the cessation of pyrolysis. It also explains why the gas source in Simulation 2 must be volumetric and mobile: using only the surface rate would strongly underestimate production in the transition zone after the first seconds.

2.3.7 Axial Dependence on the Hot Inner Surface

The axial coordinate is measured from the lower plane of the meshed slice, $y = 0.9144$ m, over its 50 mm length. In every one-percent axial band, the following figures retain the maximum over nodes on the hot inner surface of `phe0`. This restriction avoids replacing a band with no surface node by a cold node located deeper in the material. Values are obtained with DPF's native elemental-nodal averaging and then by merging duplicate global nodes at DMP boundaries. Eighty-three of the 100 bands are sampled directly; the other 17 are clearly hatched and interpolated between their neighbours.

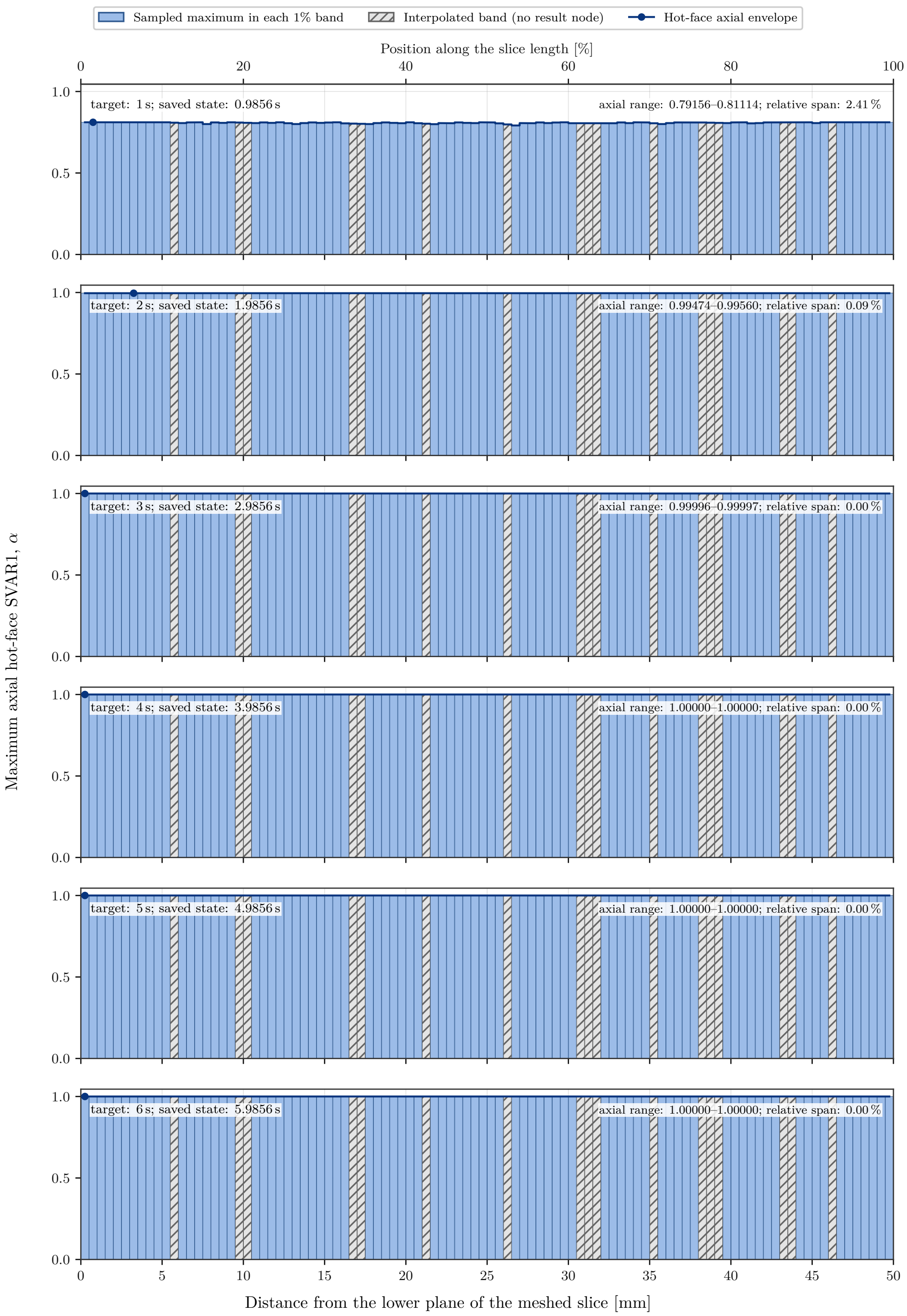

Figure 6. Axial dependence of `SVAR1` on the hot inner surface of `phe0`, measured from the lower plane of the meshed slice. All six panels use the same $0 \leq \alpha \leq 1$ scale.

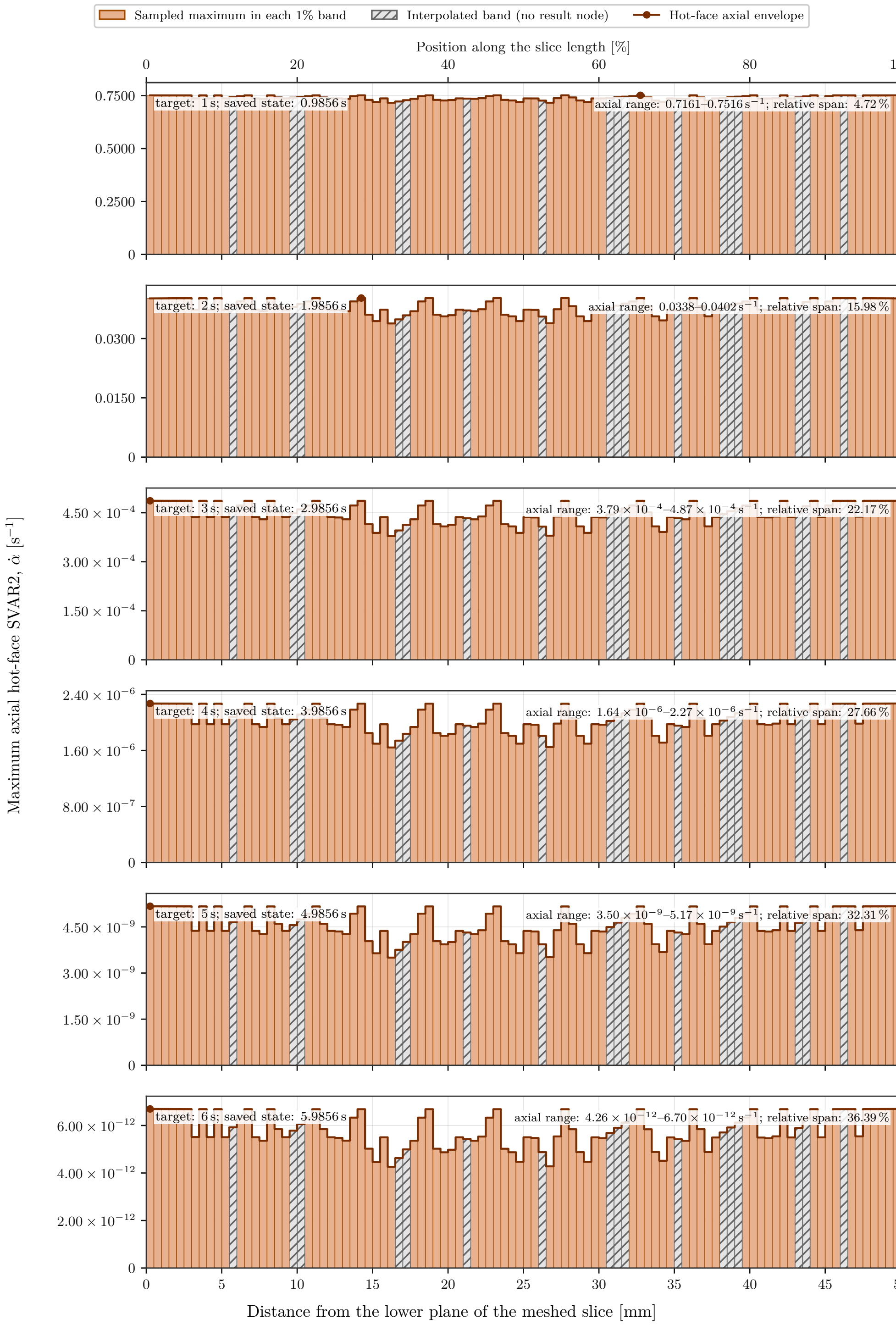

Figure 7. Axial dependence of `SVAR2` on the hot inner surface of `phe0`. The vertical scale is independent in every panel so that the small late rates remain visible; absolute values, not relative bar heights between panels, must be compared.

Figure 6 shows nearly uniform surface conversion. At 0.9856 s, α ranges from 0.79156 to 0.81114, a 2.41% relative span. At 1.9856 s, the range narrows to 0.99474–0.99560; at 2.9856 s, it is 0.999959–0.999968. The hot face is numerically fully converted from the state near 4 s. This means "char fully formed" according to the state variable, not "material removed".

Figure 7 provides the complementary information. The hot-face rate is still $0.7161 - -0.7516\,\mathrm{s}^{-1}$ near 1 s, but drops to $0.033\,82 - -0.040\,25\,\mathrm{s}^{-1}$ near 2 s, then to $3.79\times10^{-4} - -4.87\times10^{-4}\,\mathrm{s}^{-1}$ near 3 s. It falls below $2.3\times10^{-6}\,\mathrm{s}^{-1}$ near 4 s and is practically zero afterwards. This sharp drop is the expected consequence of the $1-\alpha$ factor: the surface has almost no virgin material left to convert, while Figure 5 confirms that the volumetric front remains active deeper in the material.

The relative axial span of `SVAR2` increases from 4.72% at 1 s to 36.39% at 6 s, but this increase is misleading if read alone: the denominator has fallen by about eleven orders of magnitude. Late variations are therefore negligible in absolute value and mostly reflect mesh, nodal smoothing, and numerical saturation. Under the uniform boundary conditions of this central slice, both figures validate the assumption of a nearly axially-uniform response. They support the future use of a much cheaper axisymmetric model for the central region, but say nothing about the ends, the nozzle, joints, or real combustion-flux variations.

2.3.8 Bracket on Conversion and Potential Outgassing

Two simple reconstructions provide an order-of-magnitude bracket without claiming a statistical uncertainty. The lower reconstruction sets $\alpha = 0.95$ between the hot face and the contour at 1.098 mm, then $\alpha = 0$ beyond it. Accounting for cylindrical geometry gives $\bar{\alpha}_{\mathrm{low}} = 0.408$. Integration of the maximum envelope of the 100 bands gives $\bar{\alpha}_{\mathrm{high}} = 0.581$. Table 9 then uses the nominal gas yield $Y_g = 0.47$.

Reconstruction	$\bar{\alpha}$	**Equivalent converted thickness**	**Potential gas**
Lower: 95% step front	0.408	1.04 mm	0.943 kg full motor; 25.8 g slice
Upper: maximum radial envelope	0.581	1.48 mm	1.340 kg full motor; 36.6 g slice

Table 9. Engineering bracket on conversion and potential gas mass at $t = 6.3856$ s.

These values assume that the profile is representative over the full circumference and length, then use the complete `phe0` mass of 4.911 kg. In the meshed model, the length is only 50 mm and the corresponding initial mass is approximately 0.134 kg. The upper reconstruction uses radial maxima and is intentionally conservative for an average. The lower reconstruction is not a mathematical bound on the 3D solution because the figure does not contain circumferential and axial minima. Finally, "potential gas mass" is the mass that would be released according to the density law; the solver does not demonstrate that this mass has actually left the solid.

2.3.9 Physical and Motor-Design Consequences

1. **Inner-liner margin.** A thickness of approximately 1.10–1.27 mm is already 80–95% converted, and the maximum `SVAR2` lies at 1.44 mm. The front remains active and is advancing toward the epoxy. The difference $2.54 - d_{\alpha_c}$ cannot be treated directly as an ablation margin, but it shows that the first layer is heavily loaded before the intended end of the firing.
2. **Critical epoxy condition.** The model predicts 453 °C in an epoxy layer that is nevertheless assumed inert and bonded. This value lies beyond the range of its thermal-property table. The model therefore cannot establish that either the `phe0-epo` or `epo-phe1` bond survives. Characterization of the epoxy, a thicker thermal barrier, or a high-temperature adhesive must be considered before the bond can be treated as safe.
3. **Aluminum protected only over the short term.** The ideal central slice keeps the aluminum near 22 °C at 6.39 s. This is favorable, but does not cover delayed heating after shutdown, thermal bridges, or nozzle and end regions. A transient cooldown phase must be added to search for the delayed maximum aluminum temperature.
4. **Non-negligible outgassing.** The `SVAR2` value implies a high local gas source. Without porous transport, the simulation cannot predict internal char pressure, blowing cooling, or gas-assisted delamination risk. These phenomena directly motivate Simulation 2.
5. **Front resolution.** Eight elements over 2.54 mm provide only three to four element layers across the main transition zone. Runs with 16 and then 24 radial layers should compare $d_{0.8}$, $d_{0.95}$, $\dot{\alpha}_{\max}$, and the epoxy temperature.

6. **Central-slice limitation.** The 50 mm mesh represents a uniform mid-length region, but not end-region flux concentrations or conduction paths. The masses in Table 9 can be scaled to the full motor only if axial uniformity is verified.

2.3.10 Comparison with Reality and Direction of the Bounds

There is currently no TGA/DSC/LFA curve for the phenolic actually manufactured, no thermocouple record from a motor firing, and no post-fire measurement of mass, char depth, or recession. It would therefore be incorrect to claim quantitative agreement with reality. The available comparison concerns the mechanisms: NASA-verified charring-ablator models include conduction, decomposition, pyrolysis-gas flow, gas-solid heat exchange, thermochemical erosion, and surface motion [4, 5, 6]. Simulation 1 represents only conduction, local kinetics, virgin/char properties, and the endothermic sink.

Assumption or omission	Missing physical effect	Likely bias
Constant T_g and h_{in} from $t = 0$	No ignition ramp or pressure decay	High tendency if the real firing has ramps; unknown direction if the actual flux or particulate radiation is higher.
No mobile pyrolysis gas	No wall blowing, advected enthalpy, or pore pressure	Often a high tendency for net penetrating heat flux, but not a one-sided bound for internal temperature.
No oxidation, erosion, or particles	No thermochemical or mechanical char consumption	Lower bound for recession and geometric mass loss.
Char retained and no elements removed	The surface never recedes	Exactly zero computed recession: a model-imposed lower bound.
Perfectly bonded contacts	No gap, delamination, or evolving resistance	High tendency for ideal heat transmission, but an overly optimistic view of adhesive integrity.
Uncalibrated properties and kinetics	Unknown grade, fibers, char yield, and multiple reactions	No defensible bias direction; this uncertainty can dominate all others.
Maximum radial profile	Envelope rather than volume average	Engineering upper bound for reconstructed mean conversion.

Table 10. Expected direction of the main biases relative to a real firing.

For *geometric recession*, the lower bound of the current model is therefore 0 mm. A simple upper bound assumes that the entire equivalent converted thickness, approximately 1.0–1.5 mm, is also stripped away; it is intentionally severe because 52–56% of the condensed mass remains as char according to the current tables. Reality may lie between these values if the char remains partly attached, but strong oxidation or mechanical erosion can invalidate this bracket. Only an instrumented firing and post-fire measurement can close it.

The validation priorities are therefore to: (i) characterize the actual phenolic and epoxy by TGA, DSC, and LFA, methods explicitly used by NASA for residual mass, heat capacity, and thermal diffusivity measurements [9]; (ii) measure $T_g(t)$, pressure, and temperatures at the interfaces and on the aluminum during a firing; (iii) measure mass, residual thickness, and char depth after the firing; (iv) extend the computation to 7 s and then through cooldown; and (v) perform sensitivity studies on h_{in}, A, E_a, ΔH_p, char properties, contacts, time step, and mesh.

3 Simulation 2: Outgassing and Element Deactivation in Mechanical

3.1 Proposed Theory and Derivation

3.1.1 Solid Mass Balance and Gas Generation

Simulation 2 retains the kinetics used in Simulation 1, but distinguishes the solid residue from the gaseous products. Over an initial volume, the loss of condensed-phase mass is the gas source:

$$\frac{\partial \rho_s}{\partial t} = -\dot{\omega}_g''', \qquad \dot{\omega}_g''' = (\rho_v - \rho_c)\dot{\alpha}. \tag{21}$$

The final gas yield of the single-reaction model is therefore

$$Y_g = \frac{\rho_v - \rho_c}{\rho_v} = 1 - \frac{\rho_c}{\rho_v}. \tag{22}$$

With the current tables, Y_g ranges from 0.44 to 0.48. A nominal value of $Y_g = 0.47$ is consistent with the model, but must be confirmed using the residual mass measured by TGA.

In a complete porous-ablator model, conservation of gas mass is written as

$$\frac{\partial(\phi\rho_g)}{\partial t} + \boldsymbol{\nabla}\cdot(\rho_g \boldsymbol{u}_g) = \dot{\omega}_g''', \tag{23}$$

where ϕ is the porosity and $\boldsymbol{u}_g$ is the superficial velocity. For slow flow through the char, Darcy's law and the ideal-gas law give

$$\boldsymbol{u}_g = -\frac{K(\alpha, T)}{\mu_g}\boldsymbol{\nabla} p_g, \qquad \rho_g = \frac{p_g M_g}{R T_g}. \tag{24}$$

Ablative-material response codes such as CHAR solve coupled forms of these mass and energy balances [5, 6, 4]. A standard thermal element in Mechanical, however, has no pressure degree of freedom. Consequently, `UserMatTh` cannot solve Equation (23) by itself. Remaining within Mechanical therefore requires either a substantially more complex user-defined multiphysics element or a reduced-order model.

3.1.2 Reduced Radial Closure Compatible with Mechanical

The proposed model assumes that the gases generated in `phe0` escape rapidly toward the hot wall and that their accumulation is negligible. In cylindrical axisymmetry, the gas mass flux reaching the surface at radius r_s is then obtained directly by integrating the source:

$$\boxed{\dot{m}_g''(r_s) = \frac{1}{r_s}\int_{r_s}^{R_0} \dot{\omega}_g'''(r)\, r \,\mathrm{d}r,} \tag{25}$$

where R_0 is the outer boundary of `phe0`. This expression follows from the balance $2\pi r_s L \dot{m}_g'' = \int_{r_s}^{R_0} 2\pi r L \dot{\omega}_g''' \,\mathrm{d}r$. It conserves total gas mass even when pressure is not solved explicitly.

The gas carries sensible enthalpy. A minimal local closure adds the following term to the solid energy equation:

$$\dot{q}_{m,g} = -\frac{\dot{\omega}_g'''}{\rho_s} c_{p,g}(T - T_{\mathrm{ref}}). \tag{26}$$

This term must be used only if ΔH_p does not already include this enthalpy; otherwise it would be counted twice. A more faithful formulation adds $-\boldsymbol{\nabla}\cdot(\dot{m}_g'' h_g)$ to the energy balance, as is done in material-response models [4, 5].

3.1.3 Reduction of Convective Heat Flux by Blowing

Outgassing into the chamber produces a blowing flow that opposes heat transfer. For a steady, planar gas film with constant properties, the source-free energy equation is

$$\dot{m}_g'' c_{p,g}\frac{\mathrm{d}T}{\mathrm{d}y} = k_g \frac{\mathrm{d}^2 T}{\mathrm{d}y^2}. \tag{27}$$

Integrating between the wall and the outer edge of the film, and expressing the film thickness in terms of the no-blowing coefficient $h_0 = k_g/\delta$, gives the classical correction in the form

$$B_h = \frac{\dot{m}_g'' c_{p,g}}{h_0}, \qquad F(B_h) = \frac{B_h}{\exp(B_h) - 1}, \tag{28}$$

$$\boxed{q''_{\text{conv}} = h_0 F(B_h)(T_g - T_s).} \tag{29}$$

Because $F(0) = 1$ and $F(B_h) < 1$ for $B_h > 0$, the model recovers the current convection condition in the absence of outgassing and progressively reduces the heat flux as gas production increases. This law is an engineering closure; it does not replace a boundary-layer calculation and must be calibrated against chamber pressure, propellant composition, and test data.

3.1.4 Surface Balance, Recession, and Element-Deactivation Criterion

Pyrolysis produces char; it does not automatically remove that char. Recession requires a second mechanism—oxidation, sublimation, or mechanical erosion— and a surface energy balance:

$$q''_{\text{conv}} + q''_{\text{rad}} = q''_{\text{cond}} + \dot{m}''_g \Delta h_g + \dot{m}''_{\text{abl}} H_{\text{eff}}. \tag{30}$$

In the absence of sufficient chemical data, the following bounded, energy-controlled law is proposed:

$$\dot{m}''_{\text{abl}} = \max\left[0, \frac{q''_{\text{conv}} + q''_{\text{rad}} - q''_{\text{cond}} - \dot{m}''_g \Delta h_g}{H_{\text{eff}}}\right], \qquad \dot{s} = \frac{\dot{m}''_{\text{abl}}}{\rho_c}. \tag{31}$$

The recession depth is $s(t) = \int_0^t \dot{s}\,\mathrm{d}t$. The `phe0` elements are arranged in radial bands; a band is deactivated only when s exceeds its cumulative thickness. The criterion is therefore the integrated recession, possibly combined with $\alpha \geq 0.99$, but never $\alpha = 1$ alone.

In MAPDL, `EKILL` multiplies an element's contribution by a near-zero factor; the element remains in the mesh. A deactivated thermal element no longer contributes heat capacity or applied loads, and its state variables are reset when it is killed [2, 3]. Any residual mass and energy must therefore be recorded before `EKILL` and included in the balance. After each band is deactivated, the hot-side convection and surface elements must be transferred to the newly exposed face. With only eight layers through `phe0`, recession would advance in increments of approximately 0.318 mm. A credible recession study should use 20 to 40 radial bands, or demonstrate that the existing increment is sufficiently small.

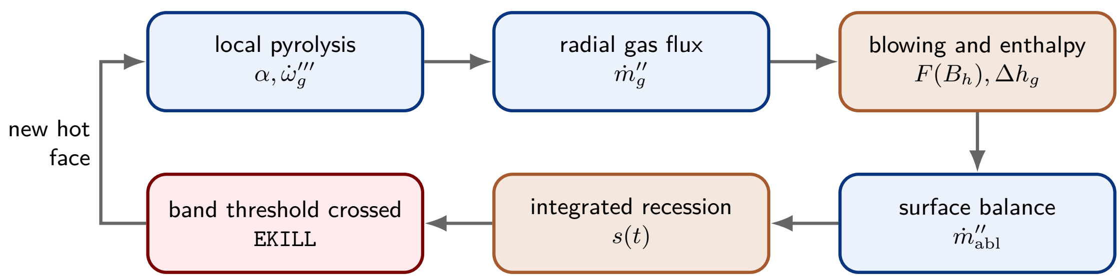

Figure 8. Proposed physical sequence for Simulation 2 in Mechanical. Element deactivation is driven by recession, not directly by SVAR1.

3.1.5 Proposed Initial Values and Sensitivity Studies

Table 11 provides an initial parameter set for prototyping, not for qualification. The permeability values are taken from the CHAR PICA model and are only surrogates for a dense phenolic [6]. Phenolic resin products include, among others, H_2, water vapor, and aromatic hydrocarbons, and depend on heating rate [8]; representing them with a single pseudo-species is therefore a major simplification.

Quantity	Proposed nominal	Sensitivity range	Rationale / required measurement
Gas yield Y_g	0.47	0.44–0.50	Derived from the current virgin/char densities using Equation (22); replace with TGA data.
Virgin porosity ϕ_v	0.13	0.05–0.20	Derived from $1 - \rho_v/\rho_{\text{skeletal}}$, with $\rho_{\text{skeletal}} = 1430\,\text{kg}\,\text{m}^{-3}$, a historical phenolic–nylon value [7].
Char porosity ϕ_c	0.55	0.45–0.65	Same calculation using $\rho_c = 600 - -700\,\text{kg}\,\text{m}^{-3}$; measure by pycnometry/porosimetry.
Virgin permeability K_v	10^{-12} m^2	10^{-14}–10^{-11} m^2	PICA "Model A" value used as a surrogate [6]; not transferable without testing.
Char permeability K_c	10^{-10} m^2	10^{-12}–10^{-9} m^2	Same source; logarithmic interpolation with respect to α is recommended.
Gas viscosity μ_g	4×10^{-5} Pa s	2–6 $\times 10^{-5}$ Pa s	Provisional hot pseudo-gas value; a mixture calculation is required.
Molar mass M_g	$0.024\,\text{kg}\,\text{mol}^{-1}$	0.018–0.030	Pseudo-species spanning light gases and vapor; calculate from the measured composition.
Gas heat capacity $c_{p,g}$	$1.5\,\text{kJ}\,\text{kg}^{-1}\,\text{K}^{-1}$	1.2–2.0	Initial pseudo-gas value; use a $c_p(T,p)$ table after characterization.
Char emissivity ε_c	0.80	0.70–0.90	Historical phenolic–nylon starting point [7]; high-temperature measurement required.
Ablation enthalpy H_{eff}	$35\,\text{MJ}\,\text{kg}^{-1}$	20, 35, and $50\,\text{MJ}\,\text{kg}^{-1}$	Effective recession parameter. $50\,\text{MJ}\,\text{kg}^{-1}$ is a historical sublimation anchor [7]; calibrate against thickness loss.
Reference temperature T_{ref}	295.15 K	fixed	Consistent with the initial state.
Recession increment	$\dot{s}\Delta t \le 0.2\Delta r$	0.1–0.25 Δr	Numerical criterion to prevent several bands from being crossed in one substep.

Table 11. Proposed initial parameters for Simulation 2. They must be treated as uncertain variables until they have been characterized.

Before a complete motor simulation is attempted, three separate validation campaigns are required: (i) TGA/DSC/LFA coupon testing for decomposition, (ii) permeability and gas-pressure measurements in a heated coupon for outgassing, and (iii) an instrumented flame or motor test for h_0, blowing, and recession. The model must not compensate for an incorrect chamber boundary condition by artificially adjusting H_{eff}.

3.2 Implementation in Mechanical

3.2.1 Computational Architecture

Simulation 2 is now implemented and running with ANSYS Mechanical APDL 2026 R1.02. It reuses the validated three-dimensional mesh from Simulation 1 and the native `UserMatTh` library. The five state variables stored in `phe0` are:

$$\text{SVAR1} = \alpha, \quad \text{SVAR2} = \dot{\alpha}, \quad \text{SVAR3} = \dot{\omega}_g''', \quad \text{SVAR4} = \int \dot{\omega}_g'''\,\mathrm{d}t, \quad \text{SVAR5} = \dot{q}_{m,g}. \tag{32}$$

The first Mechanical command object configures the user material; the second drives the transient loop. Each macro-step follows the sequence

$$\texttt{SOLVE} \longrightarrow \text{gas and energy balance} \longrightarrow \text{blowing and recession update} \longrightarrow \text{multiframe restart.}$$

After the first macro-step, `ANTYPE,,REST,,,CONTINUE` explicitly resumes the last converged state. Temperature and the SVARs are therefore not recomputed from $t = 0$.

The computation targets 7 s using 140 uniform macro-steps of $\Delta t = 0.05$ s. The 2.54 mm-thick inner `phe0` layer is divided into eight radial bands, each 0.3175 mm thick. The controller uses $T_g = 2230.85\,^\circ\text{C}$, $h_0 = 1000\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, $c_{p,g} = 1600\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$, $\rho_c = 600\,\text{kg}\,\text{m}^{-3}$, and $H_{\text{eff}} = 35\,\text{MJ}\,\text{kg}^{-1}$. A band is deactivated only when all three conditions below are satisfied:

$$\bar{\alpha} \ge 0.99, \qquad \alpha_{\min} \ge 0.90, \qquad s \ge (i+1)\Delta r.$$

After the surface-effect elements have been installed, the current model contains approximately 2.24 million nodes and 7.15×10^5 elements. The inner surface contains 135,900 nodes and 43,488 surface-effect elements; the first candidate band contains 39,808 elements.

Numerical parameter	**Executed value**
Version and solver	MAPDL 2026 R1.02, nonlinear transient thermal analysis
Parallel execution	distributed memory, Intel MPI, 10 ranks, one thread per rank
Time horizon	7 s, 140 macro-steps of 0.05 s
Global mesh	2,242,327 nodes; approximately 714,676 elements after setup
Hot surface	135,900 nodes; 43,488 surface-effect elements
`phe0` bands	8 bands, each 0.3175 mm thick
Persistence	CSV history at every step and distributed restart files retained on all 10 ranks

Table 12. Executed numerical configuration for Simulation 2.

3.2.2 Persistence, Recovery, and Consistency Checks

The `simulation2_history.csv` file is closed and reopened in append mode after every macro-step. A `simulation2_state.parm` file stores the last solved time, balances, recession, and active band. Restart files remain distributed; a recovery is valid only when the ten rank-specific sets, the history, and the parameter state point to the same load step.

The first campaign was stopped cleanly after load step 15 at $t = 0.75$ s, then resumed from load step 16. The current continuation restored the result at 0.80 s and proceeded without reinitializing temperature or the SVARs. This recovery is an important robustness check because a complete computation requires several days of wall-clock time.

Recession-rate statistics are extracted read-only with DPF from the ten `file0.rth` through `file9.rth` files. For each state, the extraction checks all 135,900 inner-surface nodes and matches the result times against the history. Figures are published only after the distributed result files have stabilized.

3.3 Simulation 2 Results and Performance

3.3.1 Run Status and Numerical Performance

The results in this subsection form a *provisional snapshot* frozen on July 23, 2026. The last fully synchronized state is load step 123 at $t = 6.15$ s, or 87.9% of the target physical duration. The computation is still running, and no extrapolation over the remaining 0.85 s is used as a final result.

The observed continuation from load step 17 through load step 123 contains 107 converged macro-steps and no MAPDL errors. Of these steps, 87 converged in two equilibrium iterations and 20 in three; every step from load step 50 onward converged in two equilibrium iterations. The median wall time per macro-step is 41.7 minutes over the full continuation. It increases as the result files grow: over the latest 24 available steps, the median is 58.9 minutes, corresponding to approximately 0.051 s of simulated time per wall-clock hour. The latest ten steps each require 58 to 63 minutes.

Thermal solution time is not the only dominant cost. After every convergence, MAPDL assembles and interprets the distributed results; this phase can currently take approximately 40 minutes. The resident database allocation had to grow automatically from 1 to 4 GB. The PCG solver also reports more than 1,000 linear iterations per load step and suggests the direct SPARSE solver. Any solver change, however, must be evaluated in a separate controlled comparison using the same mesh and result-storage settings.

The continuation log contains zero errors. The `RESCONTROL,DEFINE` warning stating that the command is ignored during a restart repeats at every macro-step, with no observed loss of checkpoints. Eight initial warnings concern material-property tables evaluated slightly outside their tabulated ranges; those ranges must be extended before qualification, even though the warnings did not prevent convergence.

Performance indicator	**Provisional value**
Synchronized progress	123/140 steps; 6.15 s/7 s; 87.9%
Analyzed continuation	107 steps, from load step 17 through load step 123
Nonlinear convergence	87 steps at 2 iterations; 20 steps at 3 iterations
Median time per step, full continuation	41.7 min
Median time per step, latest 24 steps	58.9 min
Recent throughput	approximately 0.051 s simulated per wall-clock hour
Observed parallel chain	10 ANSYS ranks, 1 `mpiexec`, 1 `hydra_pmi_proxy`
Diagnostics	0 errors; non-blocking restart warnings

Table 13. Numerical performance observed before completion of the run.

3.3.2 Thermochemical Evolution and Outgassing

Table 14 lists the values recorded by the controller. Mean conversion reaches 0.50 at approximately 1.20 s, 0.90 at 2.40 s, and 0.99 at 3.50 s. The spatial minimum reaches 0.90 at 3.00 s and 0.99 at 4.00 s. At 6.15 s, the active layer is almost fully converted, with $\bar{\alpha} = 0.99999867$ and $\alpha_{\min} = 0.99999256$.

The gas flow rate first rises as pyrolysis activates, reaches $4.849\,\mathrm{g\,s^{-1}}$ at 1.90 s, then slowly decreases to $3.809\,\mathrm{g\,s^{-1}}$ at 6.15 s. Trapezoidal integration of the sampled values gives approximately 25.34 g of gas generated by that time. This mass is a balance from the reduced-order model; it is not a measurement of composition, pore pressure, or the actual instantaneous chamber flow rate.

The effective convection coefficient follows the blowing correction. It reaches a minimum of $843.75\,\mathrm{W\,m^{-2}\,K^{-1}}$ at the outgassing peak, a maximum reduction of 15.6% relative to h_0. It then recovers to $873.01\,\mathrm{W\,m^{-2}\,K^{-1}}$, still 12.7% below the no-blowing value. Despite this reduction, the maximum hot-surface temperature continues to rise and reaches 1633.23 °C at 6.15 s; a final thermal steady regime has therefore not yet been demonstrated.

t [s]	$\bar{\alpha}$	$\alpha_{\min}$	$\dot{m}_g$ [g/s]	h [W/m^2/K]	$T_{s,\max}$ [°C]	s [μm]
0.05	0.002295	0.00000002	0.293	988.95	615.97	3.80
1.00	0.415457	0.110712	4.382	856.64	1262.14	48.56
1.90	0.786302	0.523440	4.849	843.75	1366.28	81.67
3.00	0.968921	0.906589	4.527	852.58	1466.32	117.62
4.00	0.997463	0.990541	4.276	859.62	1532.90	147.31
5.00	0.999894	0.999518	4.021	866.89	1585.39	174.83
6.15	0.999999	0.999993	3.809	873.01	1633.23	204.36

Table 14. Simulation 2 snapshots. Values at 6.15 s remain provisional until the full computation is complete.

3.3.3 Virtual Recession and Deactivation Criterion

The accumulated energy-based recession reaches 204.36 µm at 6.15 s. This is 64.4% of the first-band thickness, $\Delta r = 317.5\,\mu\mathrm{m}$. The conversion criteria are already satisfied, but the geometric criterion is not; no band has therefore been deactivated. This result is physically consistent with the controller logic: complete pyrolysis is not automatically treated as material removal. The equivalent char mass associated with this virtual recession is approximately 5.14 g over the modeled surface, but it remains present in the mesh until the band threshold is crossed.

The DPF extraction available through 6.10 s confirms an almost uniform hot surface: mean temperature 1630.98 °C, standard deviation 0.058 °C, and range 1630.61–1631.37 °C. Mean recession rate decreases from $78.21\,\mu\mathrm{m\,s^{-1}}$ at 0.05 s to $24.930\,\mu\mathrm{m\,s^{-1}}$ at 6.10 s. At the latter time, the spatial standard deviation is only $0.0024\,\mu\mathrm{m\,s^{-1}}$, indicating strong spatial uniformity in the truncated model. This uniformity must not be extrapolated to ends, joints, or the full motor without a corresponding geometric model.

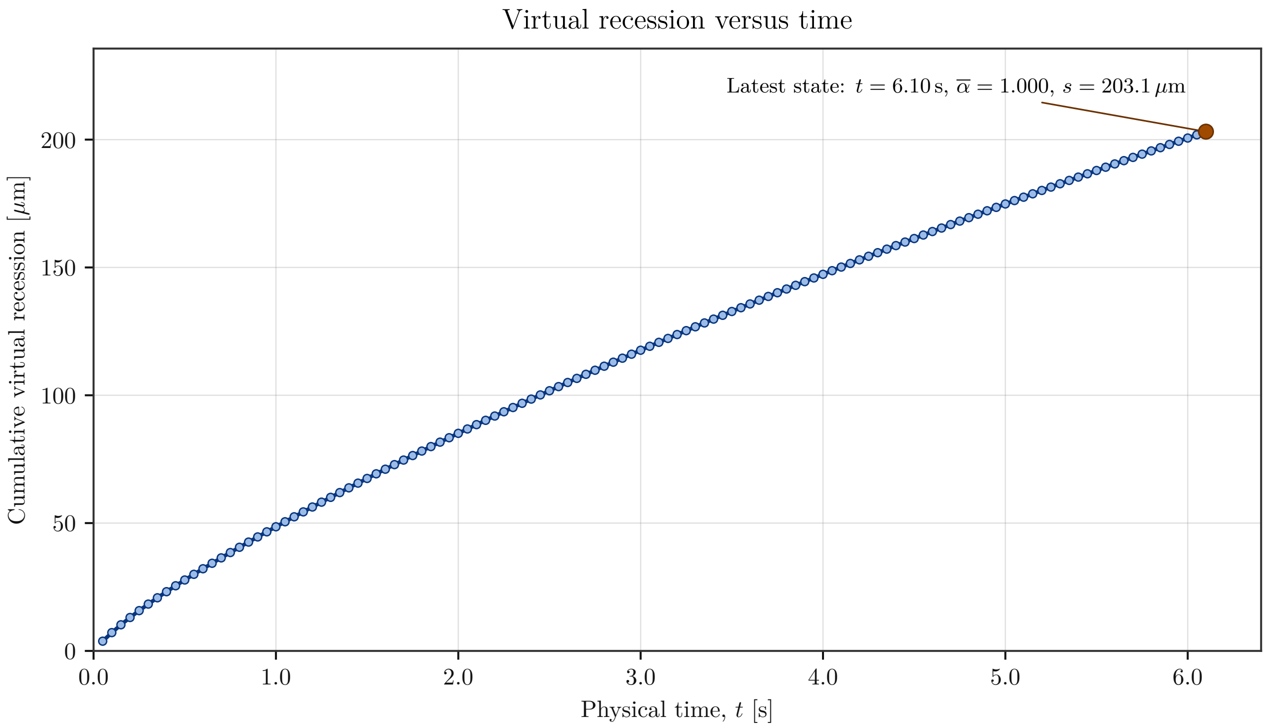

Figure 9. Accumulated virtual recession versus physical time. The curve is updated from fully synchronized states.

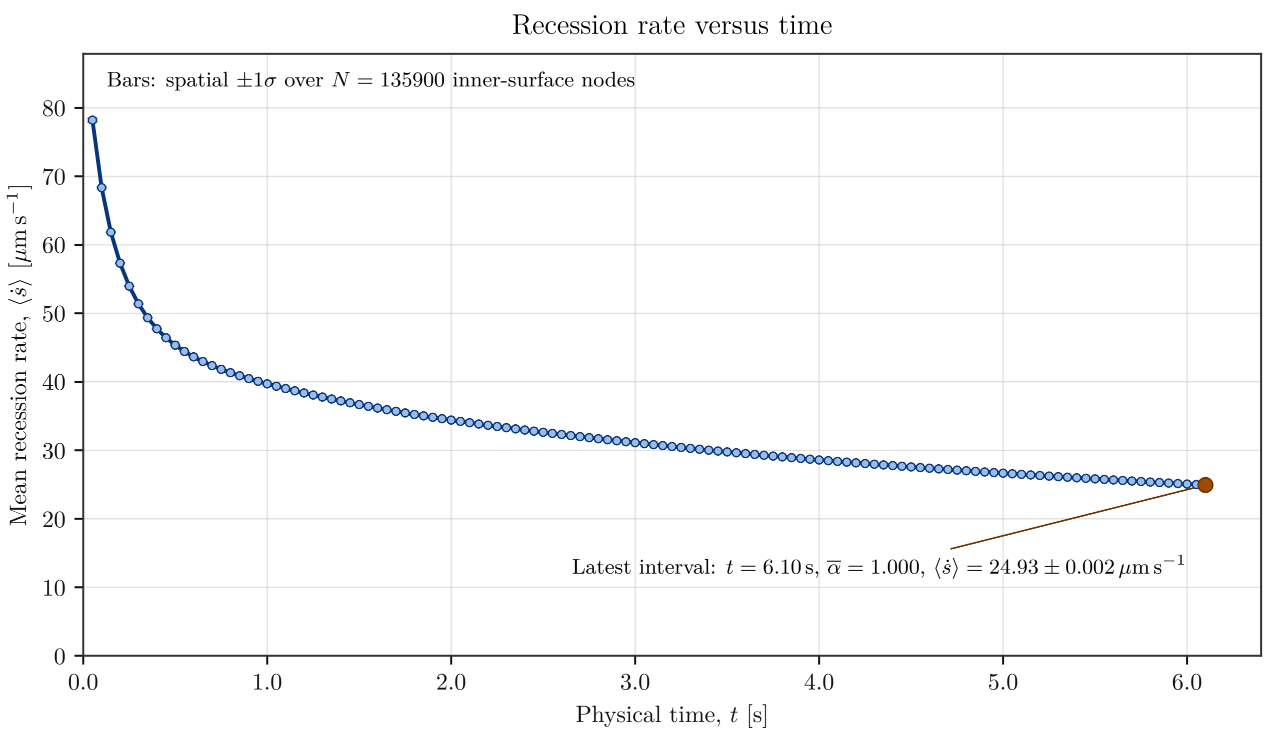

Figure 10. Mean recession rate over the 135,900 inner-surface nodes. Error bars represent spatial $\pm 1\sigma$.

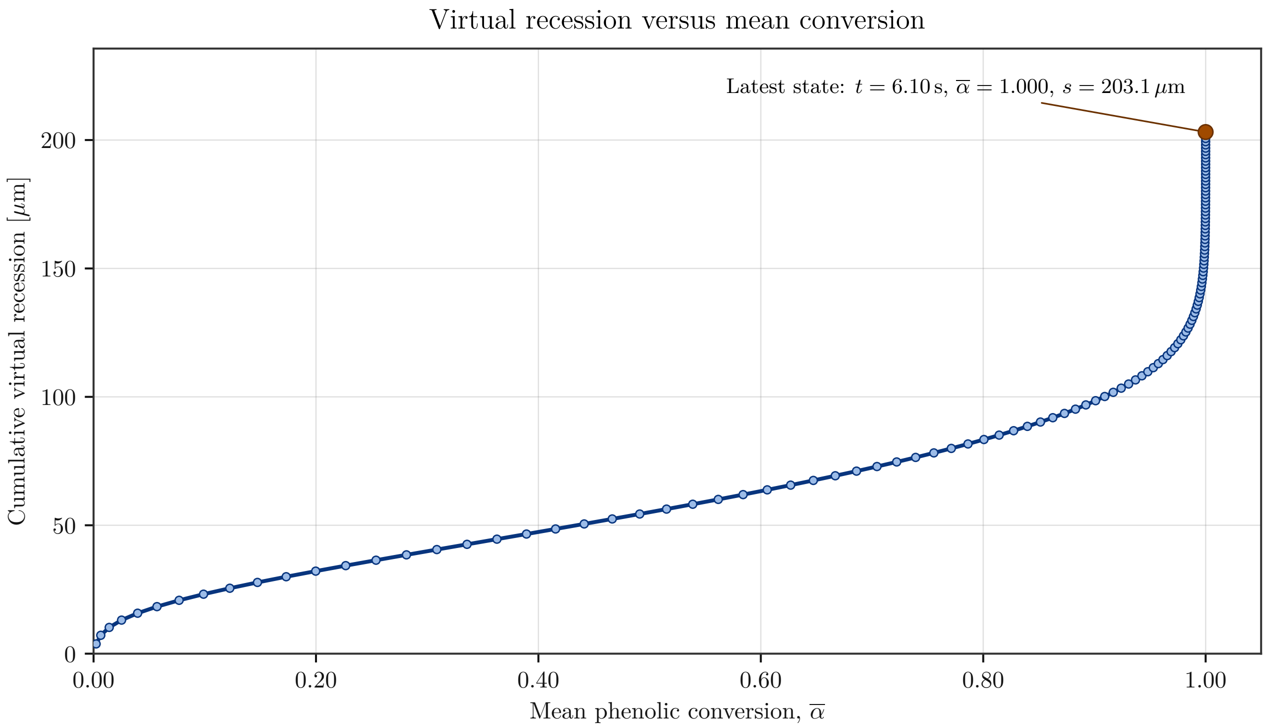

Figure 11. Virtual recession versus mean conversion. The curve shows that recession continues after $\bar{\alpha}$ has nearly saturated.

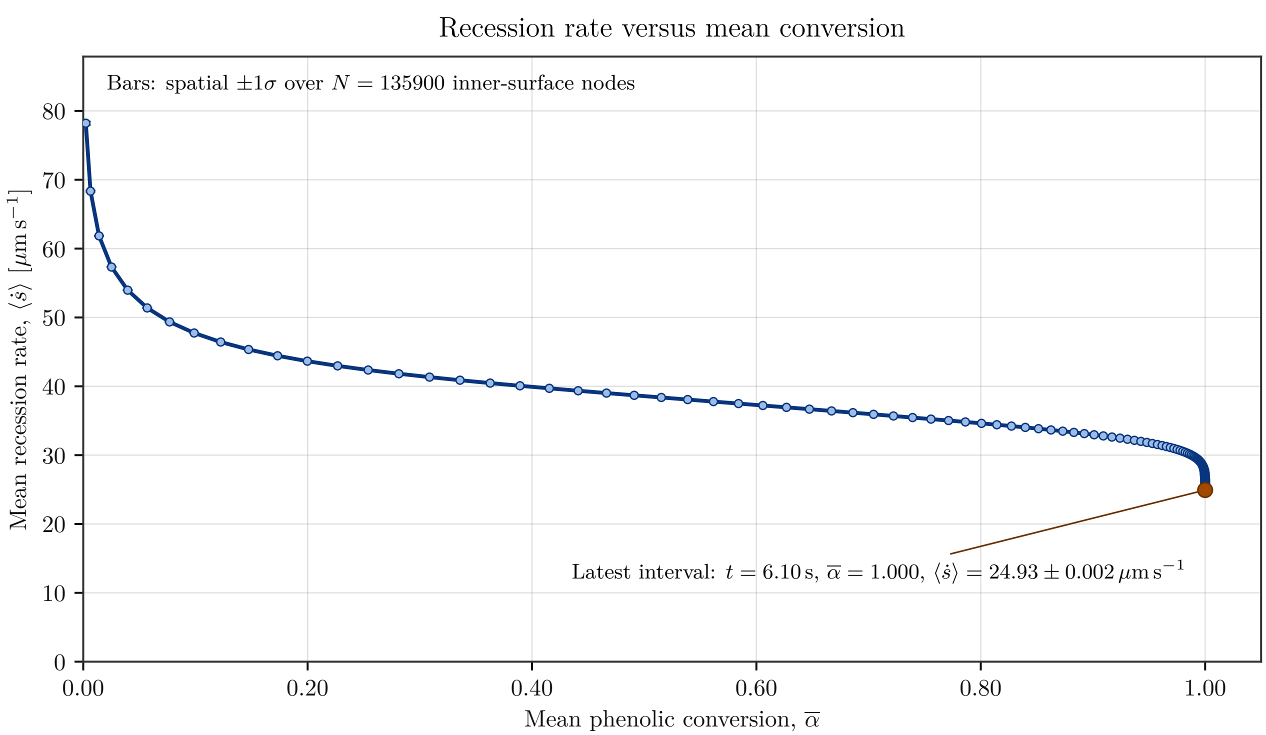

Figure 12. Recession rate versus mean conversion, with spatial $\pm 1\sigma$.

3.3.4 Spatial Conversion and Pyrolysis-Rate Profiles

To enable a direct comparison with Simulation 1, the same five profile families were reconstructed from the ten DMP partitions of Simulation 2. The saved states at $t = 1, \ldots, 6$ s are available exactly, so no temporal interpolation is used. The radial mesh, however, does not provide a node in every one of the one hundred 1% bands. Bands with no observation are hatched, and only the plotted envelope is linearly interpolated through them. This convention improves visual continuity without creating additional physical resolution.

The radial graphs also show the vertical line $d = s(t)$, where $s(t)$ is the cumulative recession computed by the controller's energy balance. No band has yet been deactivated; the line is therefore a *virtual* recession position inside the still-meshed first layer, rather than a geometric boundary that has already moved. It must not be confused with an α conversion contour.

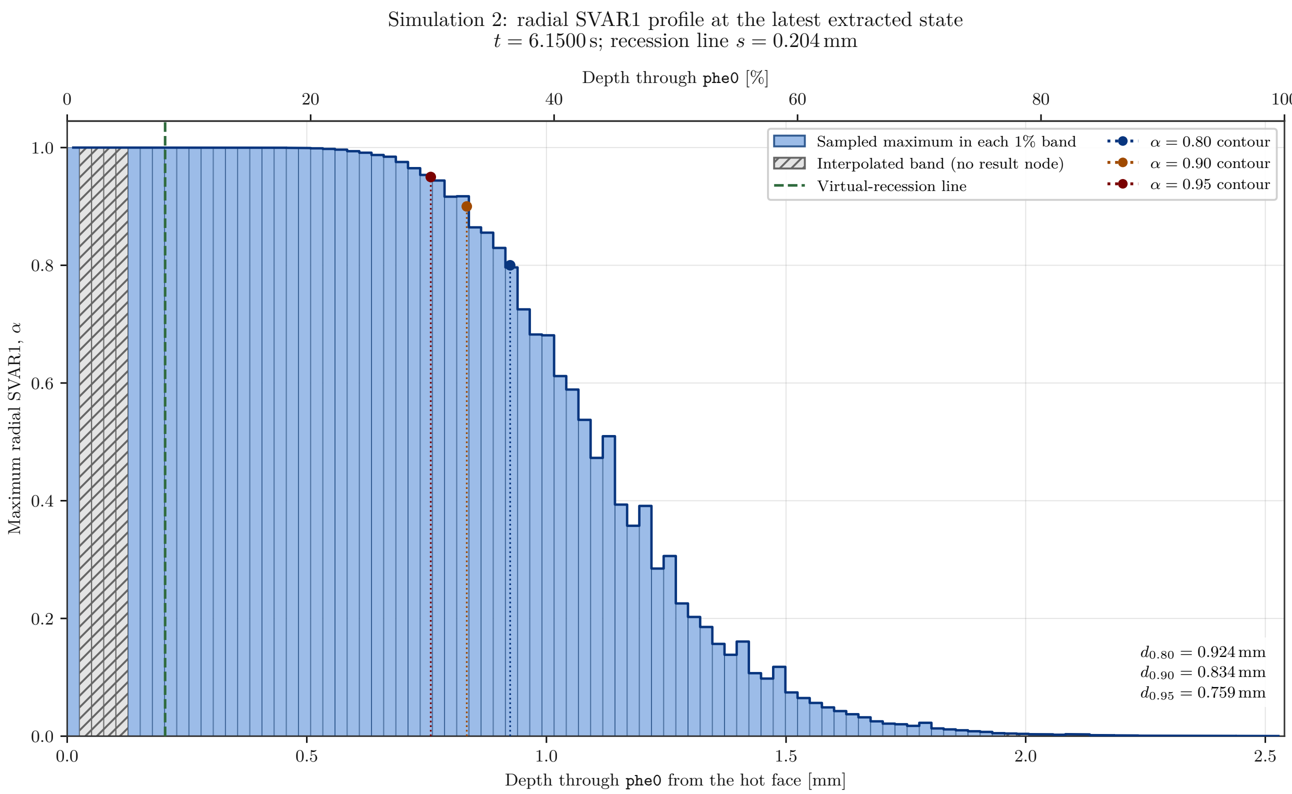

Figure 13. Radial maximum-SVAR1 profile at the latest common extracted state, including conversion contours and the virtual-recession line. Hatched bands contain no result node.

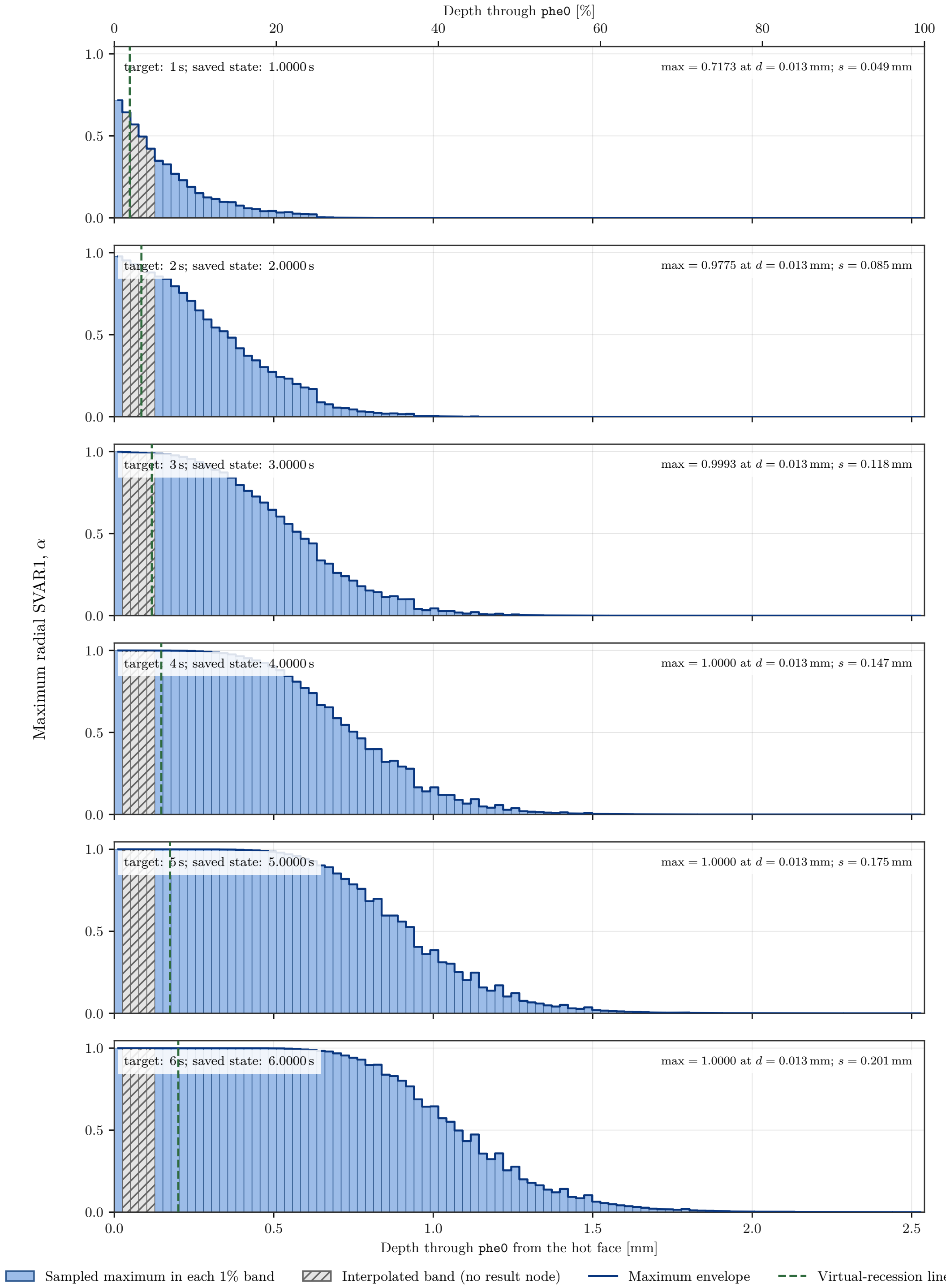

Figure 14. Evolution of the radial maximum of SVAR1 in Simulation 2. The dashed green line in each panel marks the cumulative virtual-recession position.

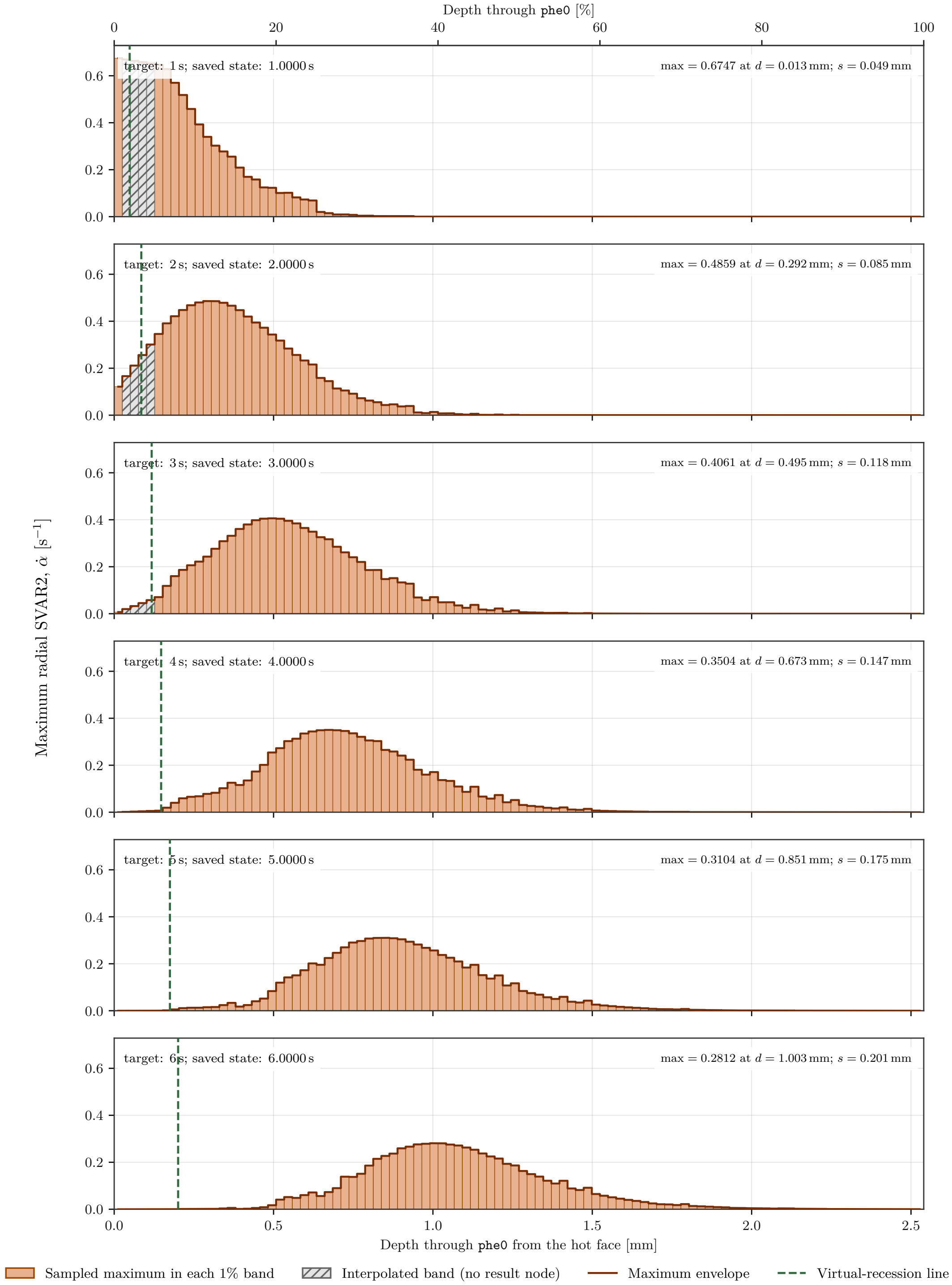

Figure 15. Evolution of the radial maximum of SVAR2. Migration of the maximum pyrolysis rate and progression of the virtual-recession line are shown on the same radial coordinate.

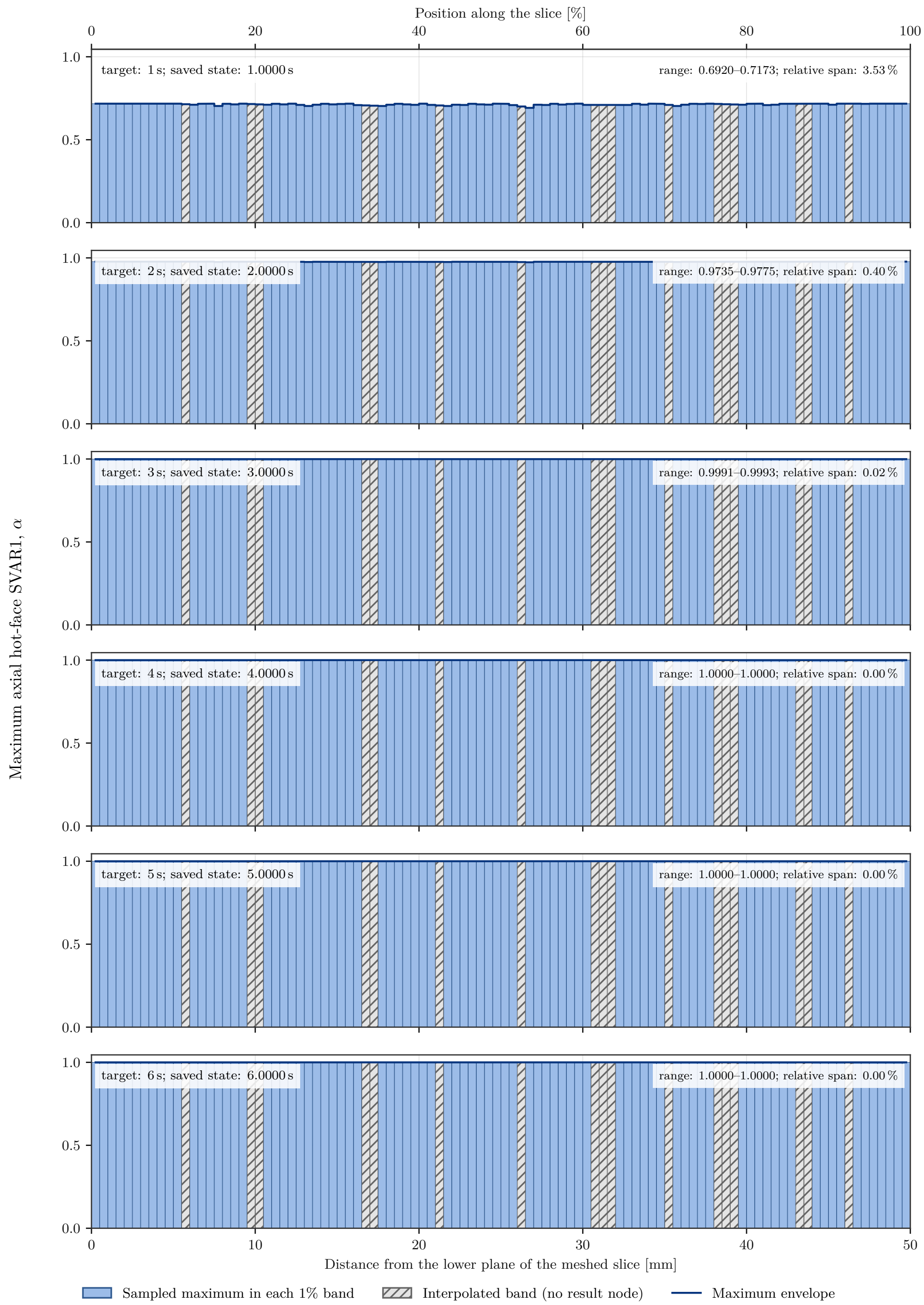

Figure 16. Axial dependence of SVAR1 on the hot inner surface in Simulation 2. No recession line is drawn on the axial axis because recession is defined along the radial coordinate.

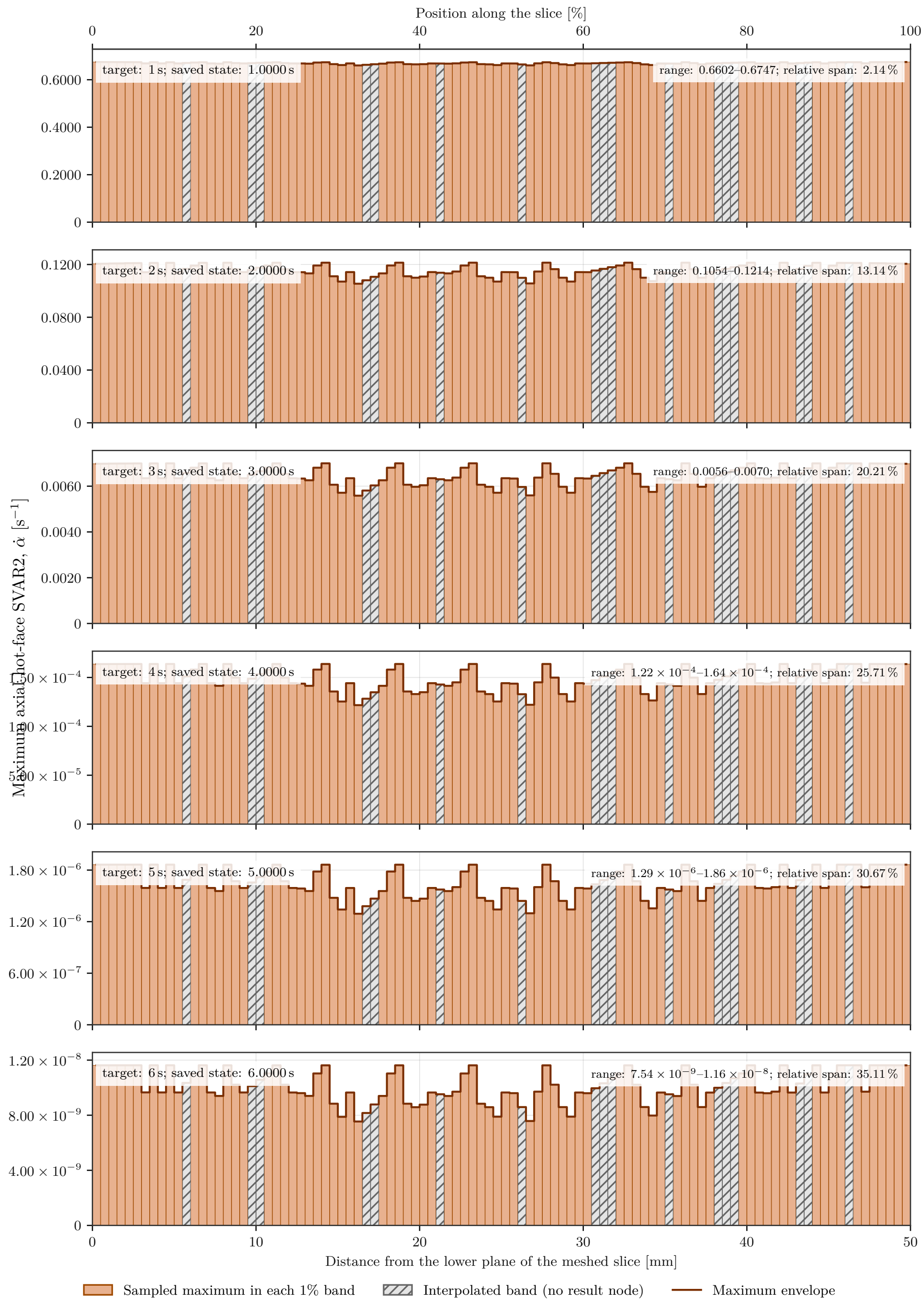

Figure 17. Axial dependence of SVAR2 on the hot inner surface in Simulation 2. An independent vertical scale in each panel reveals the small values after surface saturation.

The data retain 16 radial bands and 17 axial bands with no result node, all explicitly marked by hatching. At 1 s, the maximum radial conversion is $\alpha_{\max} = 0.7173$; it reaches numerical unity at 6 s. At the same time, the maximum pyrolysis rate decreases from $0.6747\,\mathrm{s}^{-1}$ to $0.2812\,\mathrm{s}^{-1}$ and migrates from 0.0127 mm to 1.0033 mm from the hot face. At the latest extracted state, 6.15 s, it is $0.2766\,\mathrm{s}^{-1}$ at 1.0287 mm.

The recession line progresses from 0.0486 mm at 1 s to 0.2006 mm at 6 s, and then to 0.2044 mm at 6.15 s. From 2 s onward, it remains well upstream of the SVAR2 maximum. The figures thus distinguish two different mechanisms visually: the volumetric zone where pyrolysis is most active and the cumulative material position that the energy balance would allow to recede. On the hot surface, SVAR1 becomes axially uniform at unity. Late relative variations of SVAR2 reach approximately 35%, but only for absolute values of order 10^{-8} s^{-1}, and are therefore physically negligible.

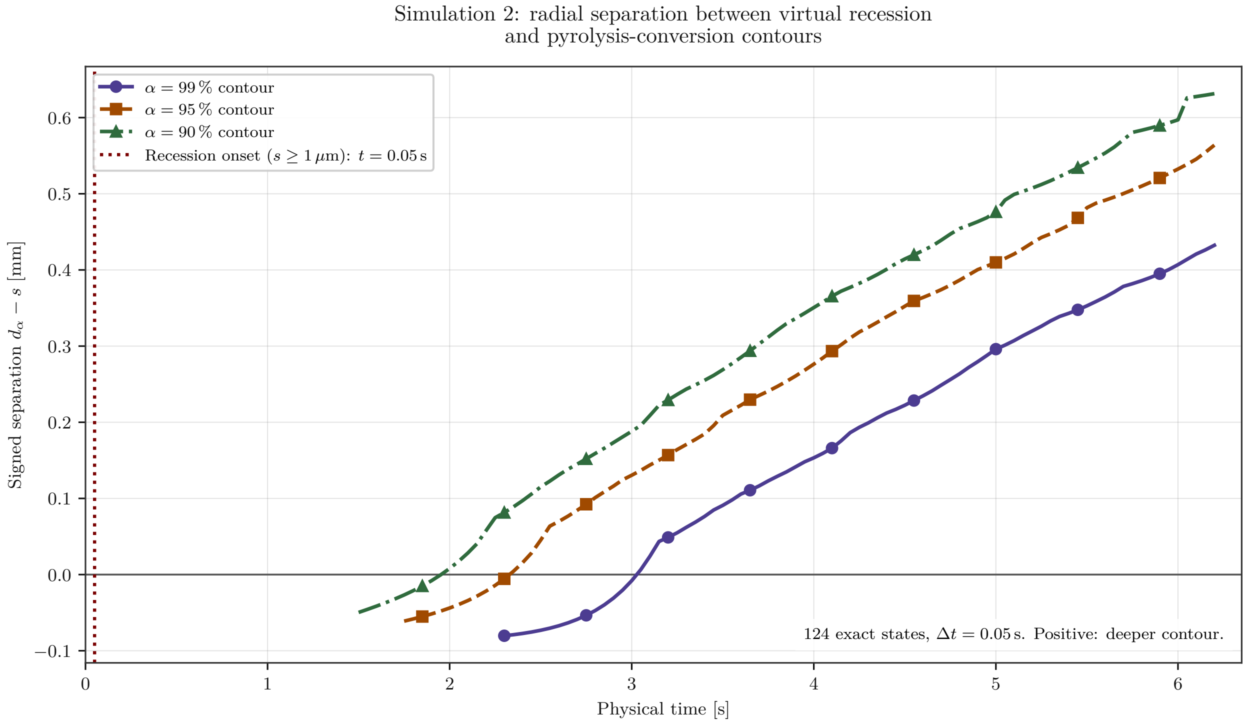

Figure 18. Signed radial separation between the virtual-recession line and the 99%, 95%, and 90% pyrolysis-conversion contours. The separation is defined as $\Delta d_\alpha = d_\alpha - s$: a positive value places the contour deeper into the wall than recession. The vertical dotted line marks the first state at which virtual recession reaches 1 µm.

The contours are obtained by spatial interpolation of the radial `SVAR1` envelope at all 124 synchronized states, spaced by 0.05 s from 0.05 s to 6.20 s; no temporal interpolation is introduced. The vertical line at 0.05 s identifies the first state for which $s \geq 1\,\mu\mathrm{m}$, with $s = 3.802\,\mu\mathrm{m}$. This micrometre-scale threshold avoids treating round-off residue as front onset while remaining far below the radial resolution of the model. A curve starts only once its conversion threshold has actually been reached, so missing observations are not replaced by artificial zeros. The 90%, 95%, and 99% contours first appear at 1.50 s, 1.75 s, and 2.30 s, respectively, initially upstream of recession with $\Delta d_\alpha = -0.0497$ mm, -0.0611 mm, and -0.0803 mm. They then pass the recession line at 2.00 s, 2.35 s, and 3.05 s. At the latest extracted state, 6.20 s, the separations reach 0.4324 mm, 0.5639 mm, and 0.6314 mm for the 99%, 95%, and 90% contours, respectively. The growth of all three separations shows that the conversion front advances through the thickness faster than virtual recession.

3.3.5 Provisional Interpretation and Remaining Work

The main result at this stage is twofold. First, the `UserMatTh`–outgassing–blowing–recession–distributed-restart chain has operated over more than one hundred macro-steps without loss of continuity. Second, with $H_{\mathrm{eff}} = 35\,\mathrm{MJ\,kg^{-1}}$, recession remains below one band at 6.15 s; the pilot therefore does not yet predict a discrete movement of the hot boundary.

The endpoint at 7 s must replace this snapshot with the final balance and determine whether the first band is crossed. The report must then be completed with: (i) closed mass and energy balances; (ii) a PCG/SPARSE comparison including memory and wall time; (iii) a mesh-convergence study using 20 to 40 radial bands; (iv) sensitivities to H_{eff}, h_0, time step, and char properties; and (v) comparison with an instrumented test. Without these steps, computed recession remains a model-performance indicator, not a qualified motor prediction.

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